

Evidential Actor Surfaces

Local Research Archive: V6–V7.1

Research snapshot: 5 September 2026

A. Reading Guide and Evidence Map

This document is a local research archive accompanying *Evidential Actor Surfaces for Physically Consistent Driving Reconstruction*. It consolidates the experiments and theory developed from WorldSim V6 through V7.1. It is intended for private inspection and research continuity, not for OpenReview submission. The scientific snapshot is the V7.1 branch at commit 1913ab0e, including the completed M43 AV2 evaluation.

The main paper follows the current surface–evidence–return formulation. This archive also retains earlier successful interfaces, rejected alternatives, diagnostic experiments, and historical visualizations. A positive result retains its original data role: development, reused cohort, independent confirmation, and synthetic capability are different kinds of evidence.

Evidence	Current role
M7 / M8	Target-set expansion and frame-supervised canonical geometry; exposed source development.
M33–M39	Producer evidence, separate anchor/child heads, and frozen categorical composition.
M22 / M28	Rigid coordinate composition and physical–visual non-interference.
M44 / M49	Exact CDF responsibility and finite attenuation identities.
M50 / M51	Frame-weighted ray-loss rejection and smooth/minimum operator diagnosis.
M43	Complete frozen AV2 comparison; hit improvement with early-return regression.
V6–V7	Earlier compilation, reliability, and selective-authority results; historical context.

Table S1. How the retained evidence enters the revised paper.

Notation and implementation IDs. EAS set expansion corresponds to M7, the frame-supervised geometry to M8, and the final categorical measure to M39 using frozen M35 anchor and M38 child heads. The unit categorical compara-

tor uses the same M8 centers and scales with unit weights. The original completion comparator in the geometry table is a point surface scored by the literal operator. These baselines are distinct. M18 is a learned query decoder; M21 is the earlier frozen unit-energy candidate; neither denotes the final evidential model.

Corrections to the previous manuscript. The revised method distinguishes the alpha-composited expected-depth surrogate from literal point-surface evaluation. M8’s pre-hit term is a hinge on expected depth; M38 separately introduces integrated pre-hit child optical thickness. M39 freezes heads trained through transmittance and then changes composition, whereas direct categorical fine-tuning was tested later as M40. The AV2 cohort is sampled at positions 0, 4, . . . , 76 from the sorted 90-log complement after excluding 60 previously consumed logs.

A read-only implementation audit also identifies a naming collision in M43: `_evaluate_bundle` first constructs literal point-surface rows, then replaces `baseline_early_count` and `baseline_hit_count` with the unit categorical counts. The `m39_surface_return` comparison remains internally matched and supports the reported M43 verdict. However, the `m8_point_surface` early/hit deltas compare a categorical baseline to a literal output. They cannot be interpreted as a matched geometry-transfer experiment. Its Chamfer pair is unaffected. We retain the immutable run and omit that mixed-operator causal attribution.

Archive structure. The next sections give reproducibility details, complete proofs, and the V6–V6.7 research lineage. They are followed by the extended V7.1/V7 method and experimental record, plus the restored V7 certificate, cohort, interval, and camera-evidence appendix. The separate source snapshot `supplement_v7_legacy.tex` is preserved in the repository.

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B. Current Method: Reproducibility Details

B.1. Data Roles and Denominators

The V7.1 corpus contains 1,004 Actors from raw and role-separated processed nuScenes producers. Candidate-bearing preparation supplies the 659 Actors used in the later geometry/evidence stages. The deterministic index-modulo-ten split yields 593 training and 66 development Actors. Parent-representation exposure predates the latter split, so the 66-Actor results are development evidence. The earlier S1 oracle uses 64 training Actors; M0/M1 use 88 source-selection Actors; B4 uses 121. The different eligibility conditions explain these different denominators.

Build observations and target observations are separated by sweep role. Both sensor origins and endpoints are transformed into each Actor’s canonical coordinates. Inputs retain only build-side evidence; disjoint target geometry supplies losses and evaluation labels. For AV2, the frozen compiler uses every third evaluation sweep with target modulo value two, at least four build frames and three target frames, and the eight configured rigid-vehicle categories.

The frozen external cohort contains 116 hazardous and 236 clear Actors. Its 466,576 hazardous and 550,076 clear rays sum to 1,016,652. The source fold contains 82,444 hazardous and 16,764 clear rays. We aggregate ray counts before division, including unmatched rays in the denominator. Chamfer averages the per-Actor symmetric nearest-neighbor distance. Frame coverage averages target points within frame, frames within Actor, and Actors within the reported group.

Hazard is a frozen Actor-level annotation derived from the supplied motion and ego interaction: short time to collision, small minimum clearance, interaction-range hard braking, or crossing. It is independent of the learned surface and is never an input or optimization target. The frozen criteria are TTC at most 3 s, clearance at most 2 m, normalized hard-brake score at least .75 within 15 m, or crossing within 5 m clearance and a 20 m forward window. Hard braking is normalized by 4 m/s^2 ; category mapping remains in the inherited compiler configuration.

B.2. Geometry Schedule

A frozen parent relocation model (M5) supplies the coordinate base. M7 expands each parent into four learned slots with ten geometry epochs followed by eight geometry/ray epochs (seed 71109, learning rate 3×10^{-4}). M8 initializes from M7 and adds equal-frame target coverage for six epochs. Its seed is 71110, hidden width 128, slot dimension 16, Actor batch size four, AdamW learning rate 10^{-4} , and weight decay 10^{-4} . Centers receive per-axis bounded 0.25 m residuals and are restricted by the configured padded cuboid. Scales are isotropic, bounded to $[0.02, 0.40]$ m, and initialized at 0.10 m. Eight nearest target points define local

planes and spacing targets.

Training uses at most 512 target rays and 512 anchors per Actor. Frame coverage samples at most 256 rays per target-origin group. The set, plane, scale, first-depth, and pre-hit hinge losses are divided by detached parent-reference values clamped below by 10^{-3} ; M8 frame distance is normalized by its frozen M7 reference. Geometry weights are 1, .25, .10, and the frame term has weight one. Ray depth has weight one and the pre-hit hinge has weight .5. Two-task PCGrad operates on the geometry-plus-frame and ray gradients.

The smooth renderer uses 48 depth samples, fixed 0.08 m support width, density multiplier two, and a residual depth 0.50 m behind the target. These fixed renderer widths differ from the learned primitive scales used by the categorical field. The training ray sampler depends on target depth as part of constructing the loss; inference samples the annotated Actor-box interval instead. Deployment emits all predicted children with all anchors and performs the fixed 0.06 m voxel deduplication. It applies no learned evidence threshold.

B.3. Evidence Schedule and Categorical Inference

M33’s sidecar aligns 297,535 observed anchors by producer identity. The heads retain canonical geometry, construction provenance, source ray/frame descriptors, projection displacement, support counts, and build free/occupied/unknown evidence. Children inherit their parent’s build evidence and child geometry. Each head uses a two-layer primitive encoder with ReLU, max-pooled Actor context, and a three-logit evidence output of hidden width 64.

M35 trains the anchor head for six epochs (seed 71135). M37 trains a separate child head with the anchor head frozen, and M38 fine-tunes it for four epochs (seed 71138) with integrated pre-hit survival. Evidence stages use Actor batches of four, learning rate 5×10^{-4} , weight decay 10^{-4} , at most 128 training rays, and 32 transmittance segments. The return objective combines segment negative log-likelihood, expected-depth L1 with weight .10, and soft evidence cross-entropy with weight .50. M38 adds child optical thickness before $d^* - .20$ m with weight one. Its fixed geometry and anchor head are unchanged.

M39 performs no optimization. It loads M8, M35, and M38, evaluates occupied weights through the same Gaussian kernel, and normalizes joint depth-primitive mass. Evaluation uses 64 samples in the Actor box padded by .15 m and a median level of .50. Observed anchors have fixed .08 m scale; children use the learned M8 scale. A small positive weight floor is used for log-domain numerical stability. The median linearly interpolates between the previous and first crossing CDF samples; at the first sample the left depth is the box entry. All evaluated M39 rays are

Table S2. Canonical geometry on 66 development Actors. Early and hit are ray percentages; CD is Actor-mean Chamfer in mm. All rows use literal point-surface returns and retain 100% of Actor states.

Method	Early ↓			CD ↓ (mm)	Hit ↑ (%)
	All	Hazard	Clear		
Original	26.74	27.80	21.56	237.67	47.67
Set expansion	24.41	24.96	21.74	234.77	49.84
+ frame coverage	25.70	26.37	22.39	231.46	50.43

Table S3. Categorical return comparison on identical geometry within each dataset. Unit: unit Gaussian energy. EAS: learned evidential weights. All values are percentages; source and AV2 are distinct cohorts.

Stratum	Early ↓		Hit ↑	
	Unit	EAS	Unit	EAS
<i>nuScenes development: 66 Actors / 99,208 rays</i>				
All	18.19	17.67	62.28	64.46
Hazardous	17.42	16.80	62.07	64.39
Clear	21.95	21.92	63.34	64.80
<i>AV2 frozen: 352 Actors / 1,016,652 rays</i>				
All	16.91	17.14	38.35	44.24
Hazardous	16.56	17.10	27.98	34.28
Clear	17.21	17.17	47.14	52.68

observable in both datasets.

B.4. Source Results and Operator Definitions

For finite point set X and unit direction v , the literal evaluator computes

$$d_X(r) = \min_{\substack{x \in X, s_x > 0 \\ \|x - o - s_x v\| \leq \epsilon}} s_x, \quad s_x = (x - o)^\top v, \quad (\text{S1})$$

with $\epsilon = .20$ m and $\min \emptyset = \infty$. This is a minimum over projected centers within a cylindrical beam; it is not an analytic Gaussian entry depth or the first intersection with a sphere. The categorical operator weights continuous Gaussian support on sampled depths and interpolates its median. The shared event tolerance is .20 m.

The source M8 absolute early rates are 25.70/26.37/22.39% for all/hazardous/clear Actors. Its clear-stratum regression is retained in the main table. The source M39 measure improves all three strata over unit energy, with 100% observability. The two source comparisons evaluate different operators and different interventions: M8 changes centers; M39 changes weights on fixed centers and scales.

B.5. Canonical Evidence Paths

Run paths share the root `/root/autodl-tmp/runs/worldsim_v71/`. For each entry below, append

`/summary.json` to read the canonical aggregate. Resolved configurations and checkpoint identities live in the same run directories.

M8: WS-V71-M8-TEMPORAL-FRAME-COVERAGE-01 / 20260904T202000Z__m8-temporal-frame-s71110-r2

M39: WS-V71-M39-CATEGORICAL-AUTHORITY-COMPOSITION-01 / 20260905T070000Z__m39-categorical-authority-r1

M43: WS-V71-M43-M39-AV2-ZERO-SHOT-01 / 20260905T091500Z__m43-m39-av2-zero-shot-r1

M44: WS-V71-M44-CATEGORICAL-SAFETY-RESPONSIBILITY-01 / 20260905T093000Z__m44-categorical-safety-responsibility-r1

M49: WS-V71-M49-VISIBILITY-SIGN-BOUNDARY-01 / 20260905T121500Z__m49-visibility-sign-boundary-r1

M50: WS-V71-M50-FRAME-BALANCED-FIRST-RETURN-01 / 20260905T060000Z__m50-frame-balanced-first-s71150-r2

M51: WS-V71-M51-SOFT-HARD-FIRST-RETURN-DIAGNOSIS-01 / 20260905T063000Z__m51-soft-hard-first-return-r1

The manuscript-local `results/eas_evidence.json` retains exact numerical extracts and task IDs. The repository's `docs/EXPERIMENTS.md` and unified `docs/RESEARCH_FAILURES.md` provide the full task chronology and engineering/scientific disposition. Historical reports under `docs/autoresearch/` supply the version-level evidence cited next.

C. Proofs and Operator Boundaries

C.1. Finite Attenuation of a Normalized Measure

Fix a ray, its sample grid, and all primitive kernels. Let $D_j > 0$ be the total kernel mass of primitive j , and $N_j(b) \in [0, D_j]$ its pre-boundary mass under the same linear interpolation as the deployed CDF. For positive weights w_j , put

$$D = \sum_j w_j D_j, \quad N = \sum_j w_j N_j, \quad (\text{S2})$$

$$F = N/D, \quad r_j = w_j D_j/D, \quad C_j = N_j/D_j.$$

Differentiating the quotient gives

$$\begin{aligned} \frac{\partial F}{\partial \log w_j} &= w_j \frac{N_j D - N D_j}{D^2} \\ &= \frac{w_j D_j}{D} \left(\frac{N_j}{D_j} - \frac{N}{D} \right) = r_j (C_j - F). \end{aligned} \quad (\text{S3})$$

After w_j is replaced by $v w_j$, $0 < v < 1$, the numerator and denominator become $N_v = N - (1 - v) w_j N_j$ and $D_v = D - (1 - v) w_j D_j$. Hence

$$F_v = \frac{F - (1 - v) r_j C_j}{1 - (1 - v) r_j}, \quad (\text{S4})$$

and subtraction yields

$$F_v - F = \frac{(1-v)r_j(F - C_j)}{1 - (1-v)r_j}. \quad (\text{S5})$$

The denominator is at least $v > 0$. Consequently, attenuation decreases F iff $C_j > F$, increases it iff $C_j < F$, and leaves it unchanged if $C_j = F$. At $v = 1$ there is no change. A zero-mass component has no effect.

For a family $\mathcal{J}$ uniformly scaled by v , replace $w_j D_j$ and $w_j N_j$ with $\sum_{j \in \mathcal{J}} w_j D_j$ and $\sum_{j \in \mathcal{J}} w_j N_j$. This proves the family identity with its own weighted conditional CDF. It does not assert magnitude-independent signs for arbitrary heterogeneous multi-component changes. Those changes can alter the balance between favorable and adverse components.

A two-component example. Take equal total contributions with $C_1 = .8$ and $C_2 = .2$. The mixture has $F = .5$. Halving the early component gives $F_v = .4$; halving the later component instead gives $F_v = .6$. Both operations reduce a positive weight. Their opposite effects follow from which part of the normalized measure is reduced.

Uniformly duplicating every component leaves the normalized categorical distribution unchanged, whereas duplicating only one family generally changes its relative contribution. This is the remaining family-count dependence tested by M40/M41. Global scale invariance should therefore not be interpreted as arbitrary discretization invariance.

C.2. Median Decisions and Component Accounting

For ordered samples, let $p_k > 0$ and $F_k = \sum_{\ell \leq k} p_\ell$. The implementation finds the first k with $F_k \geq .5$ and linearly interpolates the crossing. Within an interior positive-mass segment, the CDF is strictly increasing, so

$$\hat{d} < b \iff F(b) > .5. \quad (\text{S6})$$

Boundary equality is not an early event. At or outside the box endpoints, the implementation's endpoint convention must be applied explicitly; a universal quantile identity for distributions with atoms or plateaus would need corresponding tie rules.

Interpolation is linear in sample mass. Since $p_k = p_{k,A} + p_{k,\text{child}}$, it follows that $F(b) = F_A(b) + F_{\text{child}}(b)$ using exactly the same interpolation coefficient. M44 checks this accounting against the rendered early decision on all 99,208 valid source rays: event agreement is exact and the maximum numerical decomposition residual is 2.98×10^{-7} .

C.3. Why Expected Depth Does Not Control the Minimum

Consider target depth $d^* = 1$ and early boundary $b = .8$. A reference distribution puts all mass at $1 + \delta$, with $\delta > 0$. Its

expected-depth error is δ and its hard minimum is not early. Construct a candidate with mass

$$\varepsilon = \frac{\delta}{.5 + \delta} \quad (\text{S7})$$

at depth $.5$ and the remaining mass at $1 + \delta$. Its expected depth equals one exactly: $\varepsilon(.5) + (1 - \varepsilon)(1 + \delta) = 1$. Expected-depth error improves to zero, yet the hard minimum becomes $.5 < b$. As $\delta \rightarrow 0$, the added early mass also tends to zero. Thus arbitrarily small changes in expectation can accompany a discontinuous change in the minimum event.

This construction concerns a support-minimum operator, not the categorical median. A guarantee for a literal minimum requires a constraint on early support. A guarantee for a categorical median requires a margin on its ordered CDF. A mean smooth-loss improvement supplies neither condition by itself.

M51: M50 versus M8	Same support	Deployment
Added early rays (%)	.594	.659
Removed early rays (%)	.104	.114
Net early change (pp)	+.490	+.545
Added rays with lower smooth error (%)	38.03	36.54

Table S4. Smooth/minimum diagnosis on 99,208 rays. Same support uses limited anchors plus children without voxelization; deployment uses all anchors and the fixed .06 m voxel realization.

The mean smooth absolute error also changes from .291706 to .292153 m, so M50 exhibits both aggregate smooth regression and per-ray operator non-implication. Smooth-error and same-support hard-depth changes correlate at .038. Support realization changes 1.280% of current-model early events, but the disagreement already exists before that realization.

C.4. Rigid Equivariance and Appearance Non-Interference

Let e_P depend only on canonical physical state P , and define $Q(P, T; x) = e_P(T^{-1}x)$. For any rigid transform g ,

$$Q(P, gT; gx) = e_P((gT)^{-1}gx) = e_P(T^{-1}x) = Q(P, T; x). \quad (\text{S8})$$

For two canonical centers, rigid placement also gives $\|Rc_j + t - (Rc_k + t)\| = \|c_j - c_k\|$ because $R^\top R = I$. These statements require a common transform and unchanged canonical state.

Let visual parameters ψ own V_ψ and suppose the image optimizer receives only $\text{sg}(P)$ and updates ψ . Since the physical query reads no component of V_ψ ,

$$\frac{\partial Q(P, T; x)}{\partial \psi} = 0. \quad (\text{S9})$$

This is a structural property of the computation graph. It remains meaningful even when the physical geometry, supplied pose, or visual reconstruction is inaccurate.

C.5. Earlier Deletion and Reliability Identities

For point sets $X' \subseteq X$, Eq. S1 gives $d_{X'}(r) \geq d_X(r)$, including the empty-set convention. The early indicator can therefore only decrease under deletion. Hit recall can either increase (an early occluder is removed) or decrease (the hit is removed). This was the V7 motivation for evaluating coverage and return fidelity together.

For historical trajectory reliability, let $C_\tau = \max_j |n_{\tau j}^\top e_j| / \max(d_{\tau j}, \epsilon)$. A uniform clearance change δ satisfies

$$|C_\tau(d + \delta) - C_\tau(d)| \leq \max_j \frac{|n_{\tau j}^\top e_j| |\delta|}{\max(d_j, \epsilon) \max(d_j + \delta, \epsilon)}. \quad (\text{S10})$$

The result follows from the maximum operator and clipped denominator being 1-Lipschitz. It is a deterministic sensitivity bound.

The earlier multi-horizon event $R(H, c) = \Pr(\max_{t \leq H} C_t \leq c)$ is nondecreasing in budget c and nonincreasing in prefix horizon H . Convex interpolation between ordered budget knots preserves the first order; shared nonnegative interpolation weights across ordered horizon surfaces preserve the second. These structural orders are independent of probability calibration, which is evaluated on its own cohorts.

D. Research Lineage: V6 through V6.7

The earlier versions investigated several objects that are related to physical reconstruction but have distinct targets: reproducible editing, occupancy authority, conditional state selection, trajectory reliability, and local surface repair. This section retains their strongest results and terminal comparisons. The version closeouts and technical reports under `docs/autoresearch/worldsim_v6*` are the evidence sources.

D.1. V6: Typed World State and Exact Reuse

V6 established SceneIR ownership, deterministic world editing, provenance, lifecycle handling, and factorized verification. It separated exact reconstruction of a frozen output from the semantic or physical validity of an edit. Its final selector family studied when perception could be reused without changing the full pipeline’s outputs.

The R139 orthogonal confirmation reduced perception calls by 17.95%, with zero false reuse and 156 exact output hashes. R140 included shared sensor rendering in both full and selective costs: end-to-end reductions were 13.53% for StreetGS, 11.14% for AD-GS development, and 1.66%

for AD-GS confirmation, averaging 8.78% across conditions with zero reconstruction errors. The difference between call-count and end-to-end savings shows the effect of shared rendering cost.

The threshold-transfer alternatives remained negative: R134 had an AD-GS false negative and R136 a held-out conservative false positive. The identity-only guard was retained instead of a universal pixel-threshold claim. R140’s first two closeouts failed through Python/JSON boolean handling; the third reproduced their scientific certificate and completed the engineering record. R141 was not run. This lineage contributes state ownership and exact operational reproducibility to the current method, while its selector is outside the EAS inference path.

D.2. V6.1: Occupancy as a Physical Evidence Source

The 28-case minimum experiment separated method-visible occupancy from hidden observed-FREE evaluation. Oracle occupancy accepted 10/28 cases with zero false-safe cases and 39.83% mask yield. This established useful proposal support under the oracle.

Hunyuan’s generic generated surfaces accepted 0/6 cases in each tested arm and conflicted with measured free space. Two independent predicted-occupancy backends, Gaussian-World and IR-WM, each recovered the oracle’s 10/28 support but produced 10/10 false-safe accepted cases. The result concerns the frozen uncalibrated argmax-occupancy interfaces, not occupancy learning in general. ME-4 was unexecuted after the terminal comparison. The retained lesson for EAS is to keep unknown and contradictory observations explicit in the supervised target.

D.3. V6.2–V6.3: Constraints and Surface Risk

V6.2 introduced a constraint-aware evidential state and a single evidence-dropout recovery. With $p = .5$ feature dropout and teacher consistency, source-valid UNKNOWN fell from 82.74% to 63.85%. The four accepted legacy cases nevertheless remained the same four false-safe cases. The recovered projection had zero hard violations over 939,206 queries and full safe-OCC retention. Thus satisfying known constraints and reducing abstention did not resolve hidden free-space errors.

V6.3 moved to native surface and tail-risk supervision. Early objectives collapsed to infeasible states; a recovered checkpoint made the common comparison executable. On two development scenes, the B3 surface model increased pooled tail CVaR from .491496 to .608174 while reducing emitted occupied points from 2,298,450 to 1,047,186. Both scene-level comparisons failed. Zero known hard violations and nonzero useful output coexisted with worse hidden-surface risk. Surface-Max, Surface-CVaR, and downstream authority arms remained unexecuted. These results moti-

vate direct supervision of both the surface and the measured return event.

D.4. V6.4: Conditional Compilation at Fixed Opportunity

Native uncertainty features became useful after a full 273-dimensional risk head replaced the weaker PCA-16 representation. At nominal .4 coverage, independent calibration had zero failures and confirmation had one failure in 96 cases at .39994 realized coverage. A stratum-conditional map raised mean coverage to approximately .475 with zero failures on its separate 96-case confirmation cohort.

The strongest route-aware result uses an untouched, fixed-opportunity 96-case denominator: M1 preserved M0 mean coverage .474970 while reducing worst-ten route-conflict CVaR from .020726 to .010821 and pooled conflict density from .004945 to .002001. M1 was lower/equal/higher in 18/78/0 paired cases. Earlier selected-count ratios were kept separate because they changed the evaluated opportunity set.

The collision-critic extension failed. Initial unsafe-action recalls were .02174/0/.01087 for Real-only/naive/verified augmentation on 184 unsafe actions. Independent threshold recovery achieved .62044 recall for the verified arm and still produced two selected-policy false-safes; AUROC shifted from .71165 to .56274. The relative route-conflict result therefore remains an earlier positive experiment with its own target. It is not evidence of a learned collision controller.

D.5. V6.5: Visited-State Reliability

V6.5 changed the query to the reliability of world states visited by a given Ego trajectory over two seconds. Frozen Qmean transferred on fresh cohorts: P2V obtained Spearman .63396, unsafe AUROC .99415, and 49.25% lower selected cost. A two-parameter monotone calibration map reduced MSE by 92.80% and five-bin calibration error by 88.31% without changing ranking. P10V's fixed-action cohort retained Spearman .74023, AUROC .85878, and 33.26% lower selected cost.

The combined P10X confirmation passed five of six criteria, but selected cost decreased only 16.38% against the frozen 25% requirement. This was a terminal rejection of the direct action-selection benefit. The map, Actor false-safe, learned-admission, temporal, and smooth-tail branches remain recorded as unsuccessful alternatives. The retained positive is a trajectory-conditioned reliability evaluator.

D.6. V6.6: Actor Preservation and Local Repair

V6.6 separated Actor existence from local geometry validity. The frozen local geometry head reached AUROC/AUPRC .761644/.767165 on an independent con-

sumed legacy cohort covering six scenes. A runtime package retained 127 Actors, 581 states, and 1.62 million primitives. Fixed-budget triage reduced untreated conflict exposure by 68.40%.

Natural surface repair remained a trade-off. Exact sensor support reduced conflicts by 84.77% but retained only 39.57% of clean geometry. One-voxel support retained 61.95% of clean geometry but reduced conflicts by only 41.79%. The single allowed recovery failed the joint criterion. Separately, a corrected bounded responder passed six synthetic lead-brake cases, reducing collision steps from 306 to zero with jerk at most six. This is a synthetic capability result. Natural repair and downstream learned control were not established by that demonstration.

D.7. V6.7: Uncertainty, Task Geometry, and Reliability Surfaces

V6.7 developed the earlier paper's reliability formulation through a long sequence of query objects. The central successful factorization predicts Actor residual uncertainty from source state and supplies candidate-trajectory clearance analytically. Endpoint prediction error itself is independent of which candidate trajectory queries it; task geometry determines the decision boundary and which errors matter.

From endpoint summaries to directional uncertainty. P81 reduced fixed-50% unreliable events from 57 to 26 on ten fresh scenes. However, the P84–P94 summary, Deep Sets, attention, ordinal, continuous, Gaussian, BCE, and ensemble variants failed to stably improve trajectory-level visited-state objectives. Task-conditioned occupancy-flip classifiers also reversed across cohorts. The uncertainty-tube factorization reduced fixed-50% flips from 35 to five on the fresh P108 cohort. Directional Gaussian projection strengthened ranking, while P111 exposed deterministic clearance as a strong comparator. Subsequent work therefore compared learned uncertainty directly with geometry.

P147 confirmed a multi-horizon continuous-cost result on ten fresh scenes: all five horizons (.8/1.5/2.5/3.0/3.5 seconds) had higher rank correlation and lower selected cost. Macro Spearman increment was .17419 and selected-cost difference was $-.01777$. This is distinct from the earlier binary event and from an absolute probability estimate.

Conditional density and joint horizons. P182 modeled $\log(1 + C)$ with a five-component conditional Gaussian mixture. P183's separate ten-scene confirmation reduced integrated Brier by 28.48% and mean absolute reliability error by 69.38% over P173. P199 then coupled aligned horizon marginals through input-conditioned Gaussian dependence. On 1,846 trajectories in the ten-scene P201 cohort,

Brier improved by 17.52% and calibration error by 53.37% relative to independent marginals. The P203 beta map received same-read prospective secondary support, improving P199 Brier by 3.70% and calibration error by 49.12%.

Alternative sampling, density, and dependence branches are retained as comparisons: bootstrap/DRO, condition smoothing, heavy-tailed density, spline flows, direct Brier optimization, scene-balanced sampling, mixtures, constant or low-rank copulas, Student-t copulas, direct maximum-cost density, prefix-survival density, cumulative exposure, and learned selection. Several improved development calibration or one cohort while regressing another. Their results did not replace the independently supported density/dependence stages.

Monotone runtime distillation. P227 distilled the frozen density/dependence teacher. P228’s fresh 1,720-trajectory cohort had teacher MAE .007443, a .316% Brier improvement, and a .000020 absolute calibration increase. Its measured batched student stage took .002905 seconds versus .111554 seconds for the teacher joint stage (38.4 times at that stage, excluding shared upstream work). P234’s same-read secondary surface had MAE .007101 and zero budget/horizon monotonicity violations. Input-reduction and global continuous-family variants missed their fidelity criteria. Fixed-knot interpolation retained the discrete surface and its structural monotonicity.

Terminal selective-authority family. P346 retained a reused-development result: q90 mean coverage .328415 with maximum unsafe selected-set rate .090909 on P201 task conditions. Its source-held-out-horizon rate was .267108. The executable lattice’s held-out horizon is 2.5 seconds; V7-F16 corrects the older prose label of 3.0 seconds. P358–P361 then trained all four fixed-grid horizons and tested shared calibration, horizon calibration, one-sided BCE, and pairwise ranking. Their P201 coverage/risk pairs were .306284/.112360, .293169/.105263, .342896/.107527, and .343169/.127660. None simultaneously met coverage at least .30 and risk at most .10. P346 therefore remains a reused-development positive; the terminal grid family remains negative.

D.8. Transition to V7 and the Current Surface Object

V7 combined Actor-preserving physical compilation, separate repair/hazard inputs, and the inherited reliability surface. The non-learned AV2 compiler improved depth and Chamfer while preserving Actor state. Selective repair transferred on some cohorts, but fresh ranking reversal, sensor-opportunity shortcuts, and confidently wrong interval-stable decisions exposed its limitations. The fixed-lattice downstream audit checked how repair and trajectory

reliability compose as two independent decisions; it was an open-loop interface study.

V7.1 moved the primary prediction object to supervised canonical surface geometry and return evidence. The current model inherits explicit state ownership and disciplined denominator accounting. The old selector, task-reliability density, and synthetic responder remain useful historical results in this archive, with their full V7 tables and proof record below. They are not hidden components of EAS.

E. V7.1 Experiment Atlas

The detailed narrative and tables below preserve the positive and negative experiments. This atlas supplies navigation across the stable milestone IDs and summarizes the role of each family.

Early geometry diagnostics (M2–M4). M2 retains every relocated point and is evaluated once on 20 source-final scenes (69 Actors; 134,841 rays). Hazardous early returns decrease by 5.26% and hit recall increases by 1.21 points, but Chamfer worsens by 1.58 mm, exceeding its .5 mm tolerance. M3’s signed field then extracts no zero crossing for any of 66 development Actors: its sampled SDF remains strictly positive. M4 restores zero-boundary extraction for all 66 Actors, but hazardous early returns worsen by 97.77% relative and Chamfer by 53.48 mm. The train-only gradient audit finds negative geometry/ray cosine in eight of eleven batches, with mean $-.297$, motivating M5’s measured-conflict PCGrad intervention. These outcomes have distinct source-final, development, and train-diagnostic roles.

Continuous support (M9–M12). These studies explore Gaussian support, oriented planar Gaussians, explicitly supervised thickness, and finite planar charts. The later M13–M18 table continues this operator comparison. M9 shrinks mean child scale from .162 to .112 m. Its analytic sphere hazardous early rate improves 64.85% relative, while hazardous hit recall falls from 31.77% to 25.22%. M10’s oriented support reduces hazardous support early returns by 18.45% but loses 2.39 points of hazardous support hit recall and regresses the M8 point-surface hazardous criterion. M11 freezes centers/scales and trains exact support; hazardous early relative improvement is -1.21% . M12’s finite planar charts similarly give $-.60\%$ hazardous early improvement despite a .17-point hit gain. Both miss their hazardous support criterion. These matched-operator results motivate the query-field alternatives and the later M45–M49 categorical analysis.

Appearance capacity (M23–M27). The geometry-locked physical carrier has 309 primitives in the audited

Milestones	Question / intervention	Result retained in the archive
S1, M0–M1, B4	Continuous relocation, implicit extraction, evidential TSDF	Oracle joint gain; learned UNKNOWN/extraction collapse; TSDF completeness improves while hazardous early returns worsen.
M2–M5	Loss conflict and deployment-aligned relocation	Controlled diagnoses lead to non-deletion PCGrad relocation; M5 supplies the parent coordinate base.
M6–M8	One-center versus set targets; equal-frame coverage	One-center supervision misses the hazardous-ray criterion; four-child sets and frame coverage give source geometry gains.
M9–M12	Gaussian, oriented, supervised thin, and finite planar support	Local support changes expose early/hit trade-offs; these support operators are not the final M8 point evaluation.
M13–M18	Local field, radial patch, cells, signed queries, survival, categorical decoder	M18 passes the frozen development gates; the other operators exchange early returns for missing hits or observability.
M19–M21	Joint decoder/geometry and decoder-free energy	Joint training permits geometry/scale compensation; frozen M8 energy is retained as the unit categorical comparator.
M22–M28	Rigid scene composition and appearance capacity	SE(3) and typed non-interference hold; appearance-only optimization improves views but leaves a gap to StreetGS.
M29–M33	Tail supervision, intervals, anchor provenance, evidence identifiability	Tail/interval changes miss utility gates; exact producer replay supplies informative contradictory anchor evidence.
M34–M38	Anchor/child authority and ordered transmittance	Evidence is predictable; additive optical thickness retains front-tail errors despite supervised improvements.
M39–M42	Frozen categorical composition, joint tuning, conserved family mass, interval losses	M39 improves all source strata; subsequent tuning transfers mass or error between families/strata.
M43–M44	Frozen AV2 evaluation and exact responsibility	AV2 hit gains coexist with hazardous early regression; CDF event and primitive accounting are verified.
M45–M49	Orientation, CDF supervision, incidence, visibility, attenuation analysis	Local support/visibility variants fail their criteria; the exact normalized-mixture sign explains attenuation response.
M50–M51	Equal-frame ray loss and smooth/minimum diagnosis	Worst-frame early rate regresses; same-support analysis proves that the surrogate and hard event can disagree.

Table S5. V7.1 experiment families. IDs refer to canonical experiment records, not independent datasets. Unexecuted alternatives and pre-training engineering interruptions are excluded from scientific comparisons.

Actor. SH/opacity fitting raises held-out footprint PSNR from 15.86 to 16.94 dB. A supervised fine-surfel model reaches 17.57 dB, and the coarse–fine hierarchy reaches 17.73 dB, improving all six previously exposed held-out views over the single-carrier model. The hierarchy remains 7.62 dB below the original visual Actor. These are informative positive capacity experiments and do not change the physical surface.

Engineering versus scientific outcomes. Formal records preserve pre-training failures separately: missing parent-coordinate state, native package/config binding, frozen tensor gradients, and M43 resume/single-instance issues. Their narrow recoveries do not create additional scientific replicates. The final M43 cohort completed 20/20 logs before its aggregate was read. The present authoring task reads existing records and performs no new model training, target-data evaluation, or target-domain adaptation.

F. Legacy Validity and Reliability Formulation

Let Actor i have an immutable semantic state

$$A_i = (ID_i, X_{i,0:H}, D_i), \quad (S11)$$

containing identity, trajectory, and dimensions. Appearance primitives may be associated with A_i , but collision queries consume only an Actor-local physical surface S_i^{phys} and its known/unknown mask.

We separate two random variables. Artifactness a_i depends on sensor and geometric evidence E_i , physical surface state, and provenance:

$$a_i = P(\text{artifact} \mid E_i, S_i^{\text{phys}}, P_i). \quad (S12)$$

Hazardness h_i depends on Actor dynamics, map context, and a candidate Ego trajectory τ :

$$h_i = P(\text{hazard} \mid X_{i,0:H}, \tau, \mathcal{M}). \quad (S13)$$

The goal is paired conditional invariance, not unconditional statistical independence. For the same clean Actor, a legal safe-to-hazard intervention should preserve a_i . For the

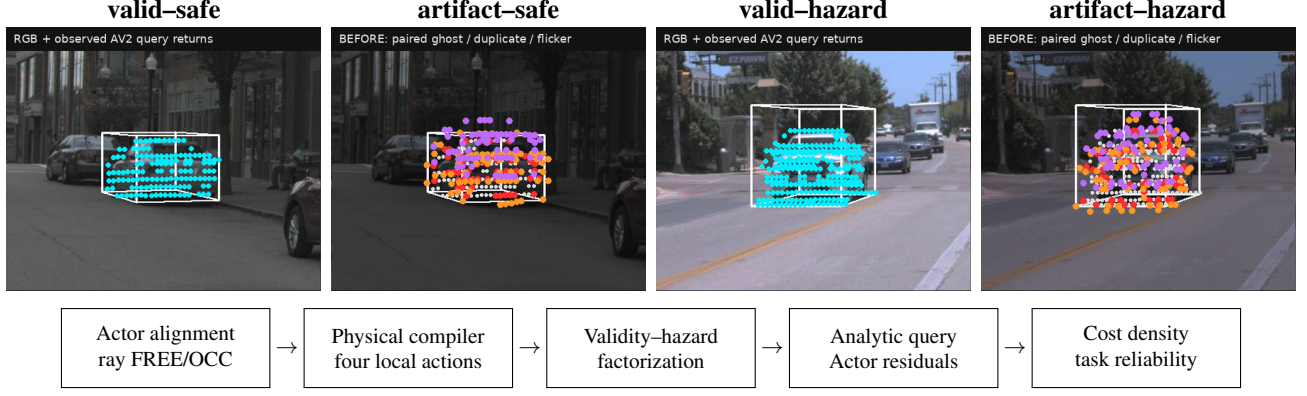


Figure S1. Validity is not difficulty. Top: a metadata-locked 2×2 teaser built from the first non-hazardous (q01-a0) and hazardous (q00-a0) frozen main cases. Within each Actor pair, only the paired synthetic artifact changes; identity, trajectory, extent, camera, and hazard label remain fixed. Bottom: HARP-3D compiles sensor evidence into an Actor-preserving physical world before granting separate task authority. These are calibrated evidence overlays, not photorealistic reconstructions.

same behavior, a clean-to-artifact intervention should preserve h_i .

For task reliability, let $n_{\tau,t}$ be the local boundary normal, $d_{i,t}(\tau)$ the signed Actor clearance, and $\epsilon_{i,t}$ the Actor residual. We define

$$C(\tau, H) = \max_{i,t \leq H} \frac{|n_{\tau,t}^\top \epsilon_{i,t}|}{\max(|d_{i,t}(\tau)|, \epsilon)}. \quad (\text{S14})$$

The runtime interface returns $R(\tau, H, b) = P(C \leq b)$ and enforces $\partial R / \partial b \geq 0$ and $\partial R / \partial H \leq 0$.

G. Extended Method and Historical Interfaces

This record includes the M18 learned decoder and earlier V7 selective/reliability interfaces. The current EAS training and inference stages are specified in Sec. B; historical variants below retain their original experiment roles.

G.1. Actor-Preserving Physical Compilation

For each Actor, we transform LiDAR endpoints, rays, and associated primitives into a canonical Actor frame,

$$x_{i,t}^{\text{actor}} = T_{\text{actor},t}^{-1} T_{\text{ego},t} T_{\text{sensor}} x_t^{\text{sensor}}. \quad (\text{S15})$$

The fused representation stores surface support, observed-free intervals, temporal counts, local normals, viewing-direction diversity, and provenance. Compilation is deterministic and local:

- KEEP: stable sensor-supported surface;
- PROJECT: a near-surface primitive contradicts observed free space and is projected to supported geometry;
- COMPLETE: a local hole has sufficient multi-frame and multi-view support;
- UNKNOWN: evidence is missing, contradictory, or insufficient.

Unknown is a first-class state and never silently becomes free. All four actions preserve $(ID_i, X_{i,0:H}, D_i)$. Appearance primitives may attach to the repaired surface using center distance, covariance-normal alignment, depth, and visibility, but physical queries do not read Gaussian opacity.

The physical certificate is visibility-conditioned. For a held-out ray $r = (o, v, d)$ and surface S , the target-facing state $\zeta(r, S)$ is one of early, hit, late, or unmatched. In the reverse direction, an output point's state $\eta(x, r)$ is contradicted, supported, occluded, or off-ray unknown. Only the contradicted state places x in observed free space; occluded and off-ray regions remain non-falsified rather than becoming occupied or free. This bidirectional partition makes the failure object inspectable while explicitly withholding a completeness certificate for unseen surface.

For the literal safety audit, define the first valid positive depth within the frozen beam tube as

$$d_S(r) = \min\{s > 0 : o + sv \in S_\epsilon\}, \quad \min \emptyset = \infty. \quad (\text{S16})$$

If a policy only deletes points, $S' \subseteq S$ implies $d_{S'}(r) \geq d_S(r)$. Hence the early predicate $\mathbf{1}[d_S(r) < d^* - \tau]$ is monotone non-increasing under deletion, as is new-early relative to a fixed query surface. Hit retention and symmetric Chamfer are not monotone under this order: deletion may expose or remove a hit, and it can move the two directed Chamfer terms in opposite directions. This is a geometric certificate for one predicate, not a collision guarantee.

Rather than collapse unknown completion to occupied or free, we can retain a set-valued first return. Let S_i^K contain only build-endpoint anchors and S_i^P additionally contain every GT-supervised completion proposal. Under the explicit coverage assumption $S_i^K \subseteq S_i^* \subseteq S_i^P$, deletion

monotonicity gives

$$d_{S_i^P}(r) \leq d_{S_i^*}(r) \leq d_{S_i^K}(r). \quad (\text{S17})$$

The endpoints represent possible and known-supported returns and are not thresholded to a single depth. Crucially, $S_i^* \subseteq S_i^P$ is an empirical completion assumption, not a consequence of the set order; we therefore report target bracketing and unbounded intervals rather than call Eq. S17 a safety certificate.

G.2. Learned Evidential Actor Surface Field

V7.1 replaces binary keep/delete on completed primitives with a supervision-native Actor-local geometry action. Every sweep is transformed by its annotated Actor pose before fusion. Build endpoints and ray-transmission intervals define occupied, free, and unknown evidence; held-out target rays never enter the input and only construct the training target T_i . Observed KEEP and matched PROJECT points form an immutable anchor set A_i .

A deployment-consistent parent model first relocates each completion seed. Because one seed per hole cannot cover a dense target, the point model attaches 4 learned slots to each parent and predicts bounded Actor-frame center residuals and isotropic support scales. The physical point surface before the training support cap and deployment voxel realization is

$$\begin{aligned} x'_{jk} &= x_j^{\text{parent}} + \Delta x_{jk}(E_i), \quad k = 1, \dots, 4, \\ S_i^\theta &= A_i \cup \{x'_{jk}\}_{j,k}. \end{aligned} \quad (\text{S18})$$

Geometry pretraining applies bidirectional Chamfer to S_i^θ and T_i , target-local point-to-plane distance, and a scale target from the target surface’s local eight-neighbor spacing. Physical fine-tuning keeps every geometry term active and adds the alpha-composited expected-depth loss and its pre-hit hinge (Sec. B),

$$\begin{aligned} \mathcal{L}_G &= \mathcal{L}_{\text{CD}} + .25\mathcal{L}_{\text{p2n}} + .1\mathcal{L}_s, \\ \mathcal{L}_P &= \mathcal{L}_G + \mathcal{L}_{\text{first}} + .5\mathcal{L}_{\text{free}}. \end{aligned} \quad (\text{S19})$$

Measured negative task gradients are projected only during this second stage. Space behind a held-out first hit is not labeled free; unknown is never a deployment deletion mask. Actor identity, trajectory, extent, and hazard state remain immutable interfaces. Image, semantic, motion, and hazard features are absent from the geometry model; a later appearance bridge may update semantics only, with stop-gradient on physical center and scale.

The historical M18 decoder separates point geometry from ray termination. Four Actor-local child centers and tangent scales per completion seed are supervised directly by target points, local point-to-plane structure, smooth expected-depth loss, its pre-hit hinge, and frame-balanced

target coverage. These losses act during training; no predicted child is deleted at deployment. Given an ordered set of query depths $d_1, \dots, d_K$ inside the Actor box, a shared 3D decoder reads the nearest child latents and relative canonical coordinates to produce logits a_k . We define

$$p_k = \frac{\exp a_k}{\sum_j \exp a_j}, \quad F_k = \sum_{j \leq k} p_j, \quad \hat{d} = \min\{d_k : F_k \geq \frac{1}{2}\}, \quad (\text{S20})$$

and supervise the native LiDAR return with

$$\begin{aligned} \mathcal{L}_{\text{return}} &= -\log p_{k^*} + \lambda_d \left| \sum_k p_k d_k - d^* \right|, \\ k^* &= \arg \min_k |d_k - d^*|. \end{aligned} \quad (\text{S21})$$

The one-hot target is built from GT before training, the decoder sees build evidence only, and the same categorical distribution is used at deployment. Normalization makes F_k monotone with unit mass and prevents multiple zero crossings; it is a representation constraint rather than a post-hoc surface filter. The present protocol assumes a return inside the Actor box and therefore does not claim no-return modeling.

To test whether the ray objective constrains the completed geometry itself, we also remove the query decoder. For immutable anchors and predicted children $S_i = \{(c_j, s_j)\}$, we define a shared Actor-canonical Gaussian energy

$$e_i(x) = \log \sum_j \exp \left(-\frac{\|x - c_j\|_2^2}{2s_j^2} \right). \quad (\text{S22})$$

The ray distribution is the softmax of $e_i(o + d_k v)$ over the same metric bins and receives the same one-hot return loss. Hence its only trainable path is through child center and scale; no visibility head can compensate for incorrect anchors. Crucially, native set, plane, scale, and frame losses remain active. The failure-directed final candidate freezes the M8 geometry before applying this energy, because joint ray fine-tuning increases scale while degrading both literal point returns and the energy’s own first returns. Observed anchors use a fixed 0.08 m support. This is a development-selected representation whose frozen external evaluation ultimately rejects transfer without target adaptation.

After the geometry-only authority ablation exposes an input-identifiability failure, we retain producer evidence per observed anchor rather than infer a deletion rule from its output. For anchor j , the build input stores construction provenance, the source query ray and frame, projection displacement, canonical hit count, temporal and view support, and continuous $m_j^{\text{build}} = (m_F, m_O, m_U)$. Disjoint held-out LiDAR rays construct m_j^{GT} with free evidence before

the endpoint, occupied evidence near the endpoint, and unknown evidence behind the endpoint or without ray support. Field names and optimization paths separate build inputs from GT targets. Neither provenance nor unknown mass is a hard keep/delete decision.

For the order-aware ablation, we replace normalization of pointwise energy by analytic Gaussian transmittance, following the volumetric construction of Vol3DGS [9]. Integrating the normalized line section of each isotropic Gaussian over ray segment $[d_k, d_{k+1}]$ gives optical thickness τ_k . We then form $T_k = \exp(-\sum_{\ell < k} \tau_\ell)$ and termination weight $w_k = T_k(1 - \exp(-\tau_k))$, as in the ordered alpha-compositing used for occupancy supervision by GaussRender [2]. Because the frozen corpus contains a target return inside the Actor box, this ablation conditions w_k on a box return and reports the residual no-return mass separately; it does not claim no-return modeling.

Rigid motion is a separate authority. For annotated pose $T_{it} = (R_{it}, t_{it})$, the deployed field is

$$e_{it}^{\text{world}}(x) = e_i(R_{it}^\top(x - t_{it})). \quad (\text{S23})$$

Thus trajectory changes where an Actor field is queried but cannot deform its canonical centers or scales. We evaluate this expression by inverse-transforming world queries, avoiding loss of local precision in large world coordinates. Static Background Gaussians are a separate scene component; neither they nor dynamic SH, opacity, or image features enter the physical query. This deployment factorization preserves the frame-balanced GT supervision used to learn e_i rather than introducing a per-frame post-hoc filter.

We make this ownership boundary explicit rather than infer it from an output audit. Inspired by object-transform factorization in scene graphs and neural simulators [7, 11], let the physical state be $P_i = (\{c_{ij}, s_{ij}\}_j)$, the read-only pose be T_{it} , and the sibling visual state V_i contain render geometry, SH, and opacity. Geometry is optimized by the GT set, local-plane, scale, frame-coverage, and smooth ray losses above. Image optimization instead receives $\text{sg}(P_i)$ and updates only V_i , where sg denotes stop-gradient. The deployed physical API is

$$Q_{it}(x; P_i, T_{it}) = \log \sum_j \exp \left[-\frac{\|R_{it}^\top(x - t_{it}) - c_{ij}\|_2^2}{2s_{ij}^2} \right], \quad (\text{S24})$$

and has no visual argument.

Proposition 1 (appearance non-interference). For fixed (P_i, T_{it}) and any visual-only update ΔV_i , $Q_{it}(x; P_i, T_{it}, V_i + \Delta V_i) = Q_{it}(x; P_i, T_{it})$ and $\partial Q_{it} / \partial V_i = 0$. This follows directly from the dependency set of Eq. S24, not from a measured post-hoc correlation. Moreover, for a common rigid world transform g , inverse querying gives $Q_{gT}(gx) = Q_T(x)$; rigid motion relocates but cannot deform the canonical field. These are

ownership and equivariance guarantees, not guarantees that the GT is complete, learned geometry or trajectory is correct, Background collision is modeled, or the resulting scene is safe.

For the appearance interface, each frozen physical center receives only the SH coefficients and opacity logit of its nearest image-trained StreetGS Gaussian with the same Actor identity. The association is computed in canonical coordinates and has no geometry gradient. Appearance center, covariance, and rotation are never copied, so visual support cannot move, resize, delete, or accept a physical primitive. The resulting sidecar is an attribute carrier rather than an image-conditioned geometry model.

For the final evidential surface measure, the categorical logit is induced directly by supervised Gaussian authority. With occupied mass o_j and kernel κ_{kj} at depth sample d_k , define

$$q(k, j) = \frac{o_j \kappa_{kj}}{\sum_{a,b} o_a \kappa_{ab}}, \quad p_k = \sum_j q(k, j). \quad (\text{S25})$$

This joint measure is invariant to whether responsibility is marginalized over depth or over primitive family. In particular, for an interior evaluation boundary $b = d_{\text{gt}} - \epsilon$ with positive bin mass and the renderer’s piecewise-linear CDF,

$$\hat{d} < b \iff F(b) > \frac{1}{2}, \quad F(b) = F_{\text{anchor}}(b) + F_{\text{child}}(b). \quad (\text{S26})$$

Equation S26 is an exact explanation of the deployed decision, not an abstention rule: $F(b)$ is never thresholded to delete primitives or cases.

The same measure exposes why learned attenuation is not monotone-safe. Let $D_j = \sum_k \kappa_{kj}$, $N_j = \sum_{d_k < b} \kappa_{kj}$ (with the renderer’s boundary interpolation), $C_j = N_j / D_j$, and $r_j = o_j D_j / \sum_a o_a D_a$. Direct differentiation yields

$$\frac{\partial F(b)}{\partial \log o_j} = r_j (C_j - F(b)). \quad (\text{S27})$$

For finite attenuation $v \in (0, 1)$ applied to a positive-mass component (or uniformly to a family), the exact change is

$$F_v(b) - F(b) = \frac{(1 - v)r_j(F(b) - C_j)}{1 - (1 - v)r_j}. \quad (\text{S28})$$

Consequently, decreasing authority lowers the pre-boundary CDF only if $C_j > F(b)$; attenuation of a later-than-mixture component increases it, and changing the attenuation magnitude cannot reverse the sign. This is a safety boundary of normalized return mixtures, not permission to use the GT-dependent sign as a deployment gate.

G.3. Validity–Hazard Factorization

The validity encoder receives ray contradiction, surface residual, temporal support, provenance, visibility, and local geometry. Its label is whether the frozen compiler’s target Chamfer is no worse than that of the untouched clean

query. The hazard encoder receives only TTC, clearance, closing speed, braking, and crossing geometry. Our factorized model uses two structurally disjoint low-capacity heads; a shared-input two-head network is the learned baseline:

$$\mathcal{L} = \text{BCE}(s_{\text{rep}}, y_{\text{rep}}) + \text{BCE}(s_{\text{haz}}, y_{\text{haz}}). \quad (\text{S29})$$

We fit features and weights on nuScenes train only. On a disjoint calibration fold, threshold q is the maximum-coverage value satisfying the frozen marginal selected-error calibration statistic

$$\hat{R}^+(q) = \frac{1 + \sum_{i=1}^n \mathbf{1}[s_i \geq q, y_i = 0]}{n + 1} \leq \alpha. \quad (\text{S30})$$

Accepted Actors use the compiled surface; abstained Actors retain the clean query. Paired input swaps measure whether either score reads the other task’s variables. Its denominator is the full calibration cohort: this statistic is not conditional error among selected Actors. A formal risk interpretation additionally requires the applicable exchangeability and risk-control assumptions; AV2 uses the frozen q only for empirical evaluation.

For a frozen stratum z , selector s_i , query/compiled Chamfer (q_i, c_i) , and introduced visible-failure indicator v_i , the complete defer-to-query system admits an exact accounting decomposition:

$$\begin{aligned} G_z &= \mathbb{E}_z[s_i(q_i - c_i)] = C_z \mathbb{E}_z[q_i - c_i \mid s_i = 1], \\ M_z &= \mathbb{E}_z[s_i v_i] = C_z \mathbb{E}_z[v_i \mid s_i = 1]. \end{aligned} \quad (\text{S31})$$

where $C_z = \mathbb{E}_z[s_i]$ is repair coverage. Population gain and introduced-failure mass are the stratum-share-weighted sums of G_z and M_z . Hence Actor-state preservation, conditional risk, and complete-system utility are distinct claims.

G.4. Actor Residuals and Boundary-Cost Density

For Actor i and horizon t , a low-capacity head predicts a 2D residual distribution $(\mu_{i,t}, \Sigma_{i,t})$. Analytic projection onto $n_{\tau,t}$ gives mean $n^\top \mu$ and variance $n^\top \Sigma n$. Equation S14 converts these residuals and deterministic clearance into a continuous trajectory cost.

We model $y = \log(1 + C)$ with the frozen P182 conditional Gaussian mixture. Its implemented inputs are the Actor residual state s_{actor} , horizon H , and deterministic clearance $d_{\text{clearance}}$,

$$p_\theta(y \mid s_{\text{actor}}, H, d_{\text{clearance}}) = \sum_k \pi_k \mathcal{N}(y; \mu_k, \sigma_k^2), \quad (\text{S32})$$

which exposes a closed-form marginal CDF for any budget. The retained P199 input-conditioned Gaussian-copula dependence model combines frozen marginals across horizons; low-rank variants were separate rejected ablations.

This factorization separates learned Actor uncertainty from analytic task geometry.

The analytic boundary also exposes deterministic geometry sensitivity. Let $a_j = |n_j^\top e_j|$ and $m_j = \max(d_j, \epsilon)$. For $C(d) = \max_j a_j / m_j$ and a uniform signed-clearance shift $d' = d + \delta$,

$$|C(d') - C(d)| \leq \max_j \frac{a_j |\delta|}{m_j m'_j} \leq \frac{a_{\max} |\delta|}{\epsilon^2}. \quad (\text{S33})$$

This follows from the max operator and clipped denominator being 1-Lipschitz; it is an error-propagation bound, not a distributional safety statement.

G.5. Monotone Runtime Surface and SceneIR

The final runtime object is a frozen grid over horizon and cost budget with monotone interpolation. Budget CDF values are non-decreasing; prefix-horizon reliability is non-increasing. Groupwise isotonic maps may calibrate predefined conditions on a disjoint source fold, but do not change these structural constraints.

The two authorities compose without implication. The repair gate first chooses the surface representation,

$$\tilde{S}_i = g_i S_i^{\text{comp}} + (1 - g_i) S_i^{\text{query}}, \quad g_i = \mathbf{1}[s_i^{\text{rep}} \geq q]. \quad (\text{S34})$$

The downstream task separately grants authority only if its multi-horizon reliability satisfies the requested budget, $a_{i,t}(c) = \mathbf{1}[R_i(t, c \mid \tilde{S}_i, \tau) \geq \rho]$. In particular, $g_i = 1 \not\Rightarrow a_{i,t}(c) = 1$: evidence can support a physical repair while Actor motion remains uncertain.

SceneIR stores Actor identities and transforms, physical surfaces, known/unknown masks, provenance, and validity fields. The AV2 adapter performs only coordinate, unit, timestamp, category, and sensor-opportunity normalization. It does not fine-tune, recalibrate, or change thresholds using AV2 quality.

H. Extended Experiments and Failure-Directed Diagnostics

The V7 tables describe the historical HARP-3D compiler. V7.1 milestones are identified explicitly by M-number; source/external roles and operator definitions follow Sec. B.

Historical V7 data and protocol. The initial V7 AV2 cohort uses every fifth UUID in the sorted list of 150 validation logs, yielding 20 quantitative and 10 qualitative logs. Training and model selection use nuScenes; the V7 AV2 evaluations use no target fine-tuning, calibration, or threshold selection. Later consumed-cohort diagnostics and disjoint confirmations retain their individual data roles, as detailed below. This initial cohort is not the V7.1 M43 cohort: M43 excludes all 60 previously consumed logs and samples 20 logs from the remaining 90 (Sec. B).

Table S6. nuScenes-only selective repair and AV2 zero-shot transfer. False repair is measured over all Actors; coverage is the accepted fraction. Chamfer is for the repair-or-clean-query output. Leak is the maximum mean prediction shift under a paired swap of the other task’s input. Each threshold is reused unchanged; “fresh” denotes independently metadata-frozen exact-once cohorts.

Dataset	Head	Repair AUROC $\uparrow$	Hazard AUROC $\uparrow$	Coverage $\uparrow$	False repair $\downarrow$	Selective CD $\downarrow$	Max leak $\downarrow$
nuScenes test	shared	64.26	91.66	11.84	3.51	0.231	28.86
nuScenes test	factorized	64.91	98.11	3.95	0.44	0.246	0.00
AV2 zero-shot	shared	77.97	92.89	55.21	3.15	0.190	23.46
AV2 zero-shot	factorized	69.96	98.05	75.55	8.99	0.182	0.00
nuScenes fresh	factorized	78.23	90.60	5.69	0.00	0.214	0.00
nuScenes fresh	sparsity-consistent	74.73	90.60	26.83	2.44	0.197	0.00
AV2 fresh	factorized	65.48	98.50	77.25	11.28	0.184	0.00
AV2 fresh	sparsity-consistent	67.62	98.50	83.94	12.43	0.178	0.00

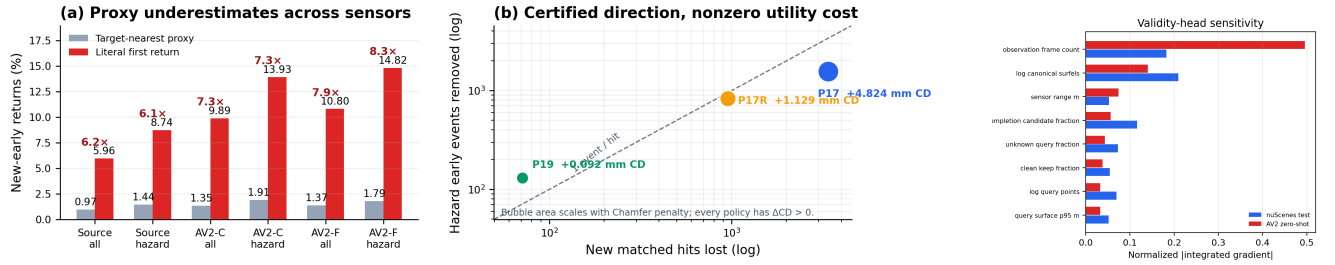


Figure S2. Physical hard evidence and boundaries. Left: target-nearest proximity undercounts literal first-return exposure on source, consumed AV2 (AV2-C), and disjoint exact-once AV2 (AV2-F). Center: deletion certifies non-increasing early returns, but every frozen policy loses hits and worsens Chamfer. Right: attribution localizes the AV2 sensor-opportunity shortcut. These are measurement, structural-direction, and model-sensitivity results—not collision or road-safety certificates.

Baselines. For physical compilation we compare naive reconstruction, a legacy validity filter, V6.7 inward-ray repair, and HARP-3D. For factorization we compare a shared encoder, a two-head shared trunk, independent encoders, and paired factorization. Reliability baselines include a scalar confidence, independent horizon classifiers, a continuous marginal density, and the full joint multi-horizon surface.

Metrics. Physical metrics include free-space violation, ray termination, point-to-surface error, LiDAR depth consistency, temporal jitter, ghost components, and surface completeness. Hazard preservation reports Actor/ID/lifecycle retention, trajectory and TTC shift, and event-count change. Factorization reports artifact/hazard AUROC and AUPRC, clean-hazard false-artifact rate, paired prediction shift, and cross-probe leakage. Reliability reports log likelihood, integrated Brier score, calibration error, Spearman correlation, monotonic violations, and run-time.

H.1. V7.1 Geometry and Query Operators

Continuous geometry oracle and learned displacement.

On train-only Actors, a bounded ray/normal displacement oracle reduces hazardous literal early returns by 5.80% while changing mean Chamfer by -3.778 mm and retain-

ing all Actor states. This establishes a feasible continuous action space without deleting completion candidates. The fixed M0 residual model does not reproduce that Pareto point on the 88-Actor source selection set. Hazardous early returns fall by 19.44%, but Chamfer changes by +48.745 mm and hit recall by -10.82 percentage points. It labels 74.51% of completion candidates unknown, so the apparent ray benefit is principally abstention rather than learned relocation. We therefore rejected M0 under the pre-registered Pareto rule and evaluate the conditionally specified implicit field next. The field further reduces hazardous early returns by 28.49%, but changes Chamfer by +120.674 mm and hit recall by -14.37 points. It extracts 0 learned surface points on every selection Actor and therefore degenerates to observed hard anchors. We rejected the field, do not tune extraction after selection, and withhold source-final and AV2 claims.

The non-learned B4 evidential TSDF exposes the opposite endpoint on 121 selection Actors. It adds 103,240 surface points, changes Chamfer by -83.961 mm and hit recall by +4.79 points, but hazardous early-return reduction is -106.53%—i.e., exposure more than doubles. Table S7 therefore shows a two-sided failure boundary rather than a winning learned surface model.

We then correct the training target rather than its de-

Table S7. Frozen V7.1 continuous-geometry evidence. Early is relative hazardous literal first-return reduction (higher is better); CD and hit are changes from each method’s query surface. S1 is a train-only oracle. M5–M7 use the same 66-Actor development fold with prior representation exposure and are not external confirmation.

Method	Early $\uparrow$	Δ CD (mm) $\downarrow$	Δ hit (pp) $\uparrow$	Decision
S1 displacement oracle	5.80%	-3.778	+2.27	feasible
M0 learned displacement	19.44%	+48.745	-10.82	reject
M1 evidential field	28.49%	+120.674	-14.37	reject
B4 evidential TSDF	-106.53%	-83.961	+4.79	opposite frontier
M5 PCGrad relocation	5.852%	-1.635	+1.429	dev pass
M6 one-to-one GT center	3.670%	-1.391	+1.077	reject
M7 GT set child surface	10.220%	-2.902	+2.173	dev pass

played output. M5 removes the UNKNOWN action and projects the measured smooth-ray/surface gradient conflict, passing all five development gates. Directly matching one Gaussian center to one nearest target point (M6) learns center, tangent-plane, and scale labels but misses the hazardous-ray gate: candidate-bearing Actors have only 17 completion seeds versus 966 target endpoints at the median. M7 instead performs fixed four-child set expansion. Its training Chamfer ratio falls from 0.986 to 0.972 before physical fine-tuning. On the same 66 exposed development Actors, it reduces hazardous/all early returns by 10.220%/8.714%, changes Chamfer by -2.902 mm, and increases hit recall by +2.173 points with complete Actor and hazard retention. M5 gives 5.852% and -1.635 mm on this fold; M6 gives 3.670% and is rejected. These are model-development results because the parent representation previously saw the fold; the frozen AV2 evaluation is reported only after all 20 logs complete.

Supervision-native geometry and first-return distribution. We next keep every generated surface and supervise the representation itself. Frame-balanced Actor-canonical seed expansion (M8) reduces hazardous early returns by 5.123%, changes Chamfer by -6.207 mm, and changes hit recall by +2.759 points. Analytic and implicit support variants then expose complementary failures (Table S8): compact support suppresses early returns but misses the surface, whereas an unconstrained direct field restores hits but creates earlier sign changes. Non-negative survival density restores monotonicity but accumulates diffuse pre-hit opacity.

The categorical model (M18) is the only frozen development candidate satisfying all seven point/ray gates. Relative to M8 it reduces hazardous/all early returns by 7.626%/4.546% and raises hazardous/all hit recall by +6.603/+5.035 points, with 100% observable rays. The clear subset remains negative (early -13.292%, hit -2.678 points), so this is not a uniform dominance or safety result. The fold was exposed to the M8 predecessor; no partial external metric is used for model choice, and the eventual

Table S8. Frozen query-representation ablation on the exposed nuScenes development fold. Early is hazardous relative reduction and hit is hazardous recall change, both versus M8. Only M18 satisfies the pre-registered early and hit gates; no row is an external result.

ID	Query representation	Early $\uparrow$	Δ hit (pp) $\uparrow$	Obs. (%)
M13	blended local planes	-76.68%	-10.57	95.90
M14	compact radial patches	+69.12%	-24.15	39.89
M15	one-sided surface cells	+35.11%	-13.54	60.41
M16	direct signed queries	-11.98%	+2.07	93.51
M17	monotone survival density	+26.52%	-26.14	96.18
M18	categorical first return	+7.63%	+6.60	100.0

Table S9. Failure-directed coupling diagnostic on the exposed nuScenes development fold. Point early is hazardous relative reduction versus the original completion (negative is worse). Energy columns are absolute hazardous early rate and hit recall. M21 reuses frozen M8 geometry and was selected by this diagnostic; it is not an external result.

ID	Representation	Point early $\uparrow$	Energy early $\downarrow$	Energy hit $\uparrow$
M18	frozen M8 + learned field	+5.12%	24.36%	57.24%
M19	joint geometry + learned field	-4.10%	24.43%	58.47%
M20	joint geometry + analytic energy	-3.25%	19.88%	57.33%
M21	frozen M8 + analytic energy	+5.12%	17.42%	62.07%

frozen M39 AV2 test rejects transfer.

Joint training exposes a stricter identifiability boundary (Table S9). M19 lets the learned field update completed children: Chamfer improves by 2.42 mm versus M8, but the point-surface hazardous early direction reverses from a 5.12% reduction to a 4.10% increase. Its apparently larger field-over-point gain therefore compensates for a degraded reference; absolute field early rate is slightly worse than M18. M20 removes the field decoder, yet predicted scale expands from 0.162 to 0.313 m on average. Its categorical NLL falls while point early returns and the analytic energy both worsen relative to their M8 initialization. We reject both fine-tuning paths.

The diagnostic nevertheless identifies a simpler candidate: frozen M8 geometry followed by the same decoder-free energy gives 17.42% hazardous early rate and 62.07% hit recall, without modifying a Gaussian after its native-3D supervision. This M21 row is selected on the exposed development fold, so it is not an independent result. We freeze its representation and the existing 20-log AV2 cohort before reading any partial quality; the later M39 external run rejects zero-shot transfer without target adaptation.

H.2. Historical V7 Compilation and Selective Interfaces

Zero-shot Actor-local surface evidence. Following the dynamic-Actor canonicalization boundary used by NeuRAD [10], we transform ego-frame AV2 LiDAR points inside each annotated cuboid into its Actor frame. Un-

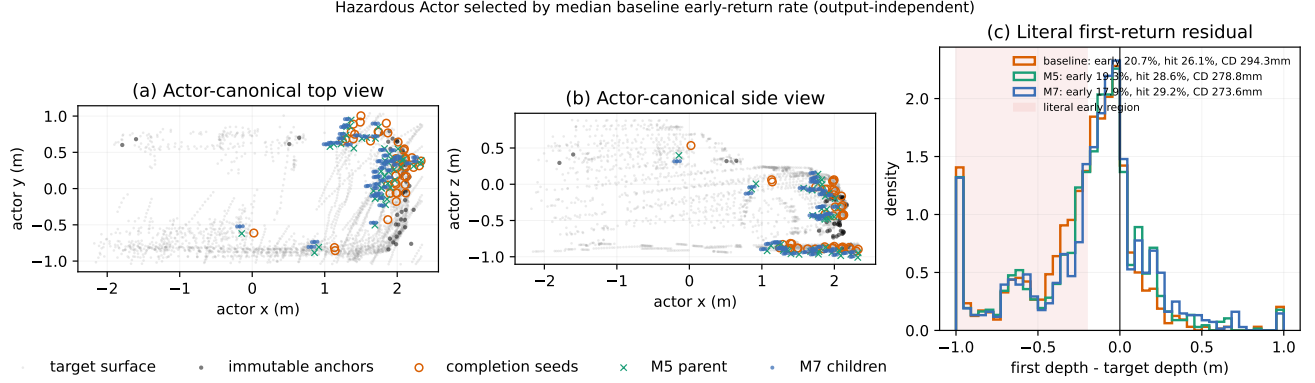


Figure S3. Supervision-native geometry on a hazardous development Actor selected by median *baseline* literal early-return rate, without consulting M7 output. Forty-seven completion seeds expand into 188 child centers while observed anchors remain fixed. For this output-independent case, baseline/M5/M7 early rates are 20.7/19.3/17.9%, hit rates are 26.1/28.6/29.2%, and Chamfer distances are 294.3/278.8/273.6 mm. Gray points are held-out GT and never model input.

AV2 Actor surface	Dist. (m) ↓	Recall (%) ↑
Single observed frame	0.864	26.16
Multi-frame canonical surfels	0.131	85.72

Table S10. Zero-shot held-out LiDAR support on the frozen 20-log AV2 quantitative cohort (1,046 Actors, including 241 hazardous Actors). The comparison uses real sensor points; paired synthetic corruptions are evaluated separately as a compiler contract.

Surface	Free (%) ↓	Recall (%) ↑	CD (m) ↓
Single-frame input	100.0	53.36	0.254
Nearest canonical	84.91	62.60	0.168
Ray-certified	0.00	59.56	0.175

Table S11. Frozen zero-shot AV2 quantitative result. Nearest-surface projection fits geometry best but violates matched-ray free space; ray-certified projection accepts a small Chamfer cost. Later held-out frames are target-only.

like appearance-oriented seed initialization, we never mirror points: every collision surfel must retain sensor and temporal support. We fuse two of every three frames at 0.12 m and reserve the remaining frames as unseen target geometry. Across 1,046 Actors and 1.075M stable surfels, fusion reduces the mean target-to-surface distance from 0.864 m to 0.131 m and raises recall at 0.20 m from 26.16% to 85.72% (Table S10).

The paired corruption atlas contains clean, observed-free ghost, duplicate-shell, temporal-flicker, and supported-hole probes without changing Actor identity, trajectory, size, or hazard. It yields 100.0% artifact recall, 0.00% clean-hazard false-artifact rate, and 100.0% Actor/hazard retention. These perfect paired scores verify the deterministic compiler contract and input separation; they are not reported as natural-artifact generalization.

Four-action physical compilation. We next freeze one single-frame query per Actor and reserve all later held-out frames as target-only geometry. The compiler reads only the build surface and query evidence. A nearest-canonical projection initially reduces Chamfer to 0.168 m, but ray-specific evaluation reveals 84.91% early termina-

tions in the paired observed-free interval. Our final operator is ray-certified: a primitive with direct LiDAR provenance projects to its matched observed termination; without one it becomes UNKNOWN. Across 433 quantitative Actors (147 hazardous), this reduces paired free-space violations from 100.0% to 0.00%, target depth error from 0.207 m to 0.154 m, and Chamfer from 0.254 m to 0.175 m, while raising ray-termination consistency from 56.94% to 64.38% (Table S11). Actor identity, trajectory, extent, and hazard labels are retained for 100.0% of Actors. The zero free-space rate is a paired contract result; depth, ray termination, and Chamfer use independent target-only LiDAR.

Frozen camera evidence and an exposed boundary. We project each pre-registered qualitative Actor back into synchronized AV2 ring-camera frames using the official calibration and motion compensation. Camera identity maximizes only the number of visible query LiDAR returns, with a fixed camera-order tie break; RGB is decoded afterward and every case is retained. All 30 cases have calibrated RGB, sparse depth, four-action, and before/after video evidence; the median/minimum visible query counts are 82/14 (Fig. S1). Importantly, the frozen set exposes a limit: al-

though mean Chamfer improves, 17.19% of all 634 Actors worsen individually. Thus P3 establishes aggregate zero-shot geometry, not a per-Actor repair certificate; this motivates a selective “repair or abstain” validity head trained and calibrated only on nuScenes.

Visibility-conditioned physical evidence. We further partition target-only LiDAR evidence with a target-nearest proximity operator into early, matched hit, late, and unmatched; reverse surface evidence is supported, contradicted by observed free space, occluded, or off-ray unknown. Unknown is never counted as free. On all 634 Actors, query-to-compiled hit recall rises from 49.69% to 69.83%, early termination falls from 3.395% to 2.868%, and visible-surface precision rises from 99.61% to 99.77% (F-score 0.663→0.822). Yet only 64.04% of Actors add no visible violation. An independent 20-log read reproduces hit 49.72→68.99%, early 3.448→2.931%, and F-score 0.663→0.815, while only 62.72% add no visible violation and 19.50% worsen in Chamfer. Provenance attribution explains the trade-off: completion supplies 95.12% of new early rays but 14.96 new hits per such ray, while only 13.49% of contradicted output points come from completion. We therefore retain the net 9,439-ray reduction, reject a universal certificate, and route Actor-level authority through abstention rather than delete completion. These frozen values localize errors around each target return; they are not literal first-return rates, which are audited below over every candidate surface point in the beam tube.

nuScenes-to-AV2 selective factorization. The strict multi-frame surface contract yields 29/56/228 Actors for nuScenes train/calibration/test; we retain the small training set rather than loosen admissibility. On nuScenes test, factorization is non-inferior to the shared head for repairability (64.91 vs. 64.26 AUROC) and raises hazard AUROC from 91.66 to 98.11, while maximum paired cross-input shift falls from 28.86% to 0.00%. The conservative threshold covers 3.95% with 0.44% population false repairs.

Without AV2 fitting, the same selector covers 75.55% of 634 Actors and reduces false repair from 16.56% under always-repair to 8.99%. Clean-query/always/selective Chamfer is 0.258/0.177/0.182m. Thus abstention costs only 5 mm mean Chamfer relative to always-repair while nearly halving false repairs. Hazard AUROC is 98.05, hazard coverage is 92.17%, and factorized cross-input shift remains zero. However, coverage changes from 8.93% on nuScenes calibration to 75.55% on AV2; Table S6 therefore supports empirical transfer, not a cross-domain conformal guarantee.

Interpretable safety envelope. We retain the model, standardizer, thresholds, and all Actor rows and evaluate

21 fixed coverage levels without refitting or recalibration. The factorized score shifts from mean/median .761/.945 on nuScenes calibration to .978/.999992 on AV2 (Wasserstein .217, KS .704), explaining the operating-point coverage jump. Integrated Gradients [8] relative to the nuScenes-training mean attributes 18.25% of absolute validity sensitivity to observation-frame count on nuScenes test but 49.53% on AV2 (Fig. S2, right). The complete fixed-threshold risk-coverage and geometry-coverage curves remain in the supplement. Structural cross-input derivatives remain exactly zero: task factorization prevents hazard leakage, but does not make validity inputs sensor invariant. We therefore preserve the empirical P4 result while rejecting the stronger interpretation of a domain-invariant selector.

We then propagate fixed 0.10 nuScenes-training-standard-deviation feature boxes through the frozen validity MLP in FP64. Only 33.33% of nominal nuScenes-test selections are robust to the full box, versus 91.23% on consumed AV2. Yet 10.76% of those AV2 robust selections are target false repairs: local threshold invariance can stably preserve a domain-shifted error. Sensor-opportunity perturbations create mean logit widths of 1.550/1.285 on nuScenes/AV2, and query-point count is the largest single-feature interval source for 83.33/95.11% of Actors. This actor-level certificate makes the shortcut inspectable, while explicitly separating network stability from repair correctness.

Fresh exact-once selector boundary. Before this read, we exclude every P4 train/calibration/test scene and freeze 20 of the remaining 106 locally available scenes by equally spaced official scene index. The 20 scenes yield 123 Actors and are read once, with no replacement or model/threshold update. Motivated by source-only LiDAR sparsity consistency [6], a pre-frozen candidate increases coverage from 5.69% to 26.83% and improves selective Chamfer from 0.214 to 0.197 m. It nevertheless lowers repair AUROC from 78.23 to 74.73 and fails the pre-registered 2-point non-inferiority margin (Table S6). A descriptive common-coverage audit follows the selective-classification AURC formulation of Zhou *et al.* [12]: candidate/P4 AURC is 0.126/0.106 (lower is better), confirming global ranking loss rather than only threshold mismatch. We retain the P4 selector; the recovery remains a negative ablation.

On 523 exact-once AV2 Actors, the order reverses: candidate/P4 AUROC is 67.62/65.48 and Chamfer 0.178/0.184 m, but false repair is 12.43/11.28%. External gates pass, yet this source/external reversal is cohort-dependent rather than universal; P4 remains primary.

Physical authority is not a visibility certificate. We exact-join the same 523 fresh AV2 Actors to the indepen-

dently frozen bidirectional visibility audit, without changing either score or threshold. P4 selects 77.25%; its Chamfer-worsening rate falls from 19.50% under always-repair to 14.60%. Selected visible-failure risk changes only from 37.28% to 36.39%, and its one-sided 95% Wilson upper bound 40.40% remains above the always-repair point risk. Safe-visible AUROC is 0.533, and abstention captures 24.62% of visible failures. Thus P4 controls its Chamfer target but does not confidence-separate visibility risk. Requiring no COMPLETE output and conjoining with P4 lowers visible failure to 20.16% at 23.71% coverage, but Chamfer worsening rises to 43.55% and hazard coverage collapses to 3.52%; observed provenance is therefore visibility evidence, not jointly valid repair authority.

A dedicated nuScenes-only visibility head confirms this mismatch rather than solving it. Safe-visible AUROC is 0.753 on source test and 0.625 on AV2. Visibility-only selection reaches 11.63% AV2 visible risk at 8.22% coverage; conjunction with P4 reaches 10.26% (95% upper 20.98%) at 7.46% coverage. Yet dual Chamfer-worsening is 35.90% and hazard coverage is 3.52%. We reject the head: low target-specific risk obtained by losing geometry and hazardous Actors is not a safety authority.

The complete defer-to-query system preserves 100.0% of Actors and separates conditional risk from population utility. P4 has 28.11% population introduced-visible failure with 0.083 m composite Chamfer gain; P6-C has 30.78% and 0.089 m, remaining an AV2-only frontier point because its fresh-nuScenes ranking reverses. P4-and-visibility has only 0.76% introduced failure but 0.0012 m gain. In contrast, the provenance policy has 4.78% introduced failure and -0.0009 m gain, so query-only strictly dominates it. Low selected risk alone therefore does not establish fallback-system utility.

Exact stratification exposes where this utility is obtained. Hazardous Actors are 27.15% of the cohort; P4 selected visible risk is 47.37%/31.00% on hazardous/clear Actors versus 46.48%/33.86% under always-repair. Only 3 of its 48 removed failures are hazardous, yet they carry 42.86% of introduced failures and 56.42% of gain. Fresh target-nearest action attribution localizes the mismatch: hazardous/clear proxy new-early rates are 1.912%/0.980% ($1.951\times$); COMPLETE causes 94.70% of new early returns and 99.94% of new hits, whereas KEEP causes 95.68% of surface contradictions. P4’s hazardous rate is $1.922\times$ ($1.006\times$ always), and P6-C is $1.001\times$. State preservation therefore does not imply hazard-stratum risk reduction or suppression of its completion-driven mechanism.

Literal first-return correction and safety-surface boundary. The preceding frozen action audit uses the target-nearest point as a proximity attribution. Following raw-ray occupancy evaluation [5], we separately re-audit

Table S12. Frozen V6.7 reliability evidence inherited without V7 refitting. P183 has 10 fresh logs; source held-out has 3,742 trajectories; P201 has 1,846 trajectories from 10 fresh logs. $\Delta B/\Delta E$ are Brier/calibration-error reductions. P346 reuses P201 and remains development evidence.

Object	Split	$\Delta B \uparrow$	$\Delta E \uparrow$	cov./unsafe $\uparrow\downarrow$
density	P183 fresh	28.48	69.38	–
joint	source held-out	16.97	71.85	–
joint	P201 fresh	17.52	53.37	–
P346	P201 reused	–	–	32.84/9.09
P346	source held-out <i>H</i>	–	–	29.42/26.71

Table S13. Frozen physical-trajectory interface on exact-match nuScenes test Actors. Cost and error are row-level 90th percentiles over retained V6.7 outcomes; FS/flip gives event counts. Physical selection is not trajectory authority.

P4 group	Actors	q90 cost $\downarrow$	q90 error $\downarrow$	FS / flip $\downarrow$
selected	5	0.283	1.577	0 / 0
abstained	83	0.102	0.693	7 / 47
geometric harm	23	0.076	0.543	0 / 4

fixed source and consumed AV2 surfaces with Eq. S16. The source total/hazard proxy rates 0.966/1.436% rise to 5.956/8.742%. P17/P17R/P19 reduce hazard exposure to 7.777/8.230/8.662%, but each worsens Chamfer. Thus the certified early direction has nonzero surface cost.

Consumed AV2 first shows hazard exposure $1.912\rightarrow 13.926\%$ ($7.29\times$). A disjoint exact-once 10-log read then independently confirms all/hazard $1.370/1.788\rightarrow 10.801/14.817\%$ ($7.89/8.29\times$). All 70,220 fresh literal new-early rays are COMPLETE; completion supplies 98.398% of 86,883 new hits, only 1.237 hits/early. Figure S2 (left and center) visualizes both findings. They transfer metric failure and provenance, not model, policy, collision, or road-safety validity.

Frozen continuous and joint reliability. We inherit the V6.7 reliability objects without V7 refitting. On 10 previously unread nuScenes logs, the five-component log-cost density reduces mean integrated Brier and marginal calibration error by 28.48% and 69.38% relative to the frozen monotone-CDF control. A conditional four-horizon copula improves joint-event Brier by 16.97% on source-held-out trajectories and by 17.52% on an independently frozen 10-log cohort; calibration-error reductions are 71.85% and 53.37% (Table S12). The downstream P346 authority reaches 32.84% q90 coverage at 9.09% unsafe rate on that consumed fresh cohort, but source held-out-horizon risk rises to 26.71%. We therefore treat density, dependence, and task authority as separate inherited evidence, not as a V7 repair effect or a formal cross-domain calibration claim.

Table S14. Retained-source 2×2 composition on 88 exact P5 Actors (32 hazardous), all retained. Coverage/cost/unsafe are over fixed two-action sets (all sets for “none”). Equal B0/B1 and B2/B3 action metrics are the intended physical/task non-interference contract, not a causal planning effect.

Arm (surface / authority)	CD ↓	cov. ↑	cost ↓	unsafe ↓
B0 query / none	0.229	100.00	0.1497	16.45
B1 HARP-3D / none	0.223	100.00	0.1497	16.45
B2 query / P346	0.229	49.43	0.0279	1.15
B3 HARP-3D / P346	0.223	49.43	0.0279	1.15

Physical repair is not trajectory authority. We exact-join P4 identities to retained V6.7 multi-horizon outcome rows through the official nuScenes scene order and DriveStudio instance tokens. Only 5 P4-train Actors from two scenes align, so we reject joint fitting rather than recycle calibration/test Actors. The frozen descriptive audit instead contains 88 test Actors and 39,285 rows (Table S13). P4 selects 5 Actors and abstains on all 23 whose compiled Chamfer is worse than the query; selected-and-harmful is therefore zero. Its 1,781 selected rows contain no false-safe or decision-flip events. Nevertheless, selected q90 motion error is 1.577 m versus 0.693 m for abstained Actors ($2.28\times$), reaching 4.246 m at 3 s. The physical gate protects representation quality but cannot replace the downstream reliability authority. These are retained source outcomes, not a causal effect of repair or an execution of the reliability model.

We also stress the analytic geometry-to-cost bound on 575,596 retained profiles under 6 fixed $\pm 0.05/0.10/0.20$ m shifts. FP64 evaluation has zero violations. At -0.20 m, full-source mean/q99 absolute cost shift is 0.0292/0.1450, while only 0.841% of rows cross signed clearance. P4-selected versus abstained mean shift is 0.00085/0.00456 ($5.38\times$ more stable) with no selected crossings. Selected Actors can therefore be motion-difficult yet geometrically stable, further separating the three axes of repairability, geometry sensitivity, and trajectory uncertainty.

Minimal composed-authority utility. Task-relevant monitoring should propagate prediction error to the queried planning cost [3], rather than treat all prediction errors alike. We therefore form a retained-source 2×2 audit over query/HARP-3D surfaces and no-task/frozen-P346 authority. The 88 exact Actors (32 hazardous) yield 1,228 fixed two-action sets at the P346 artifact’s held-out 2.5 s horizon. Frozen 90% requested reliability authorizes 49.43% of sets, reducing conditional mean actual cost from 0.1497 to 0.0279 (81.37%) and unsafe-set rate from 16.45% to 1.15%. Independently, selective HARP-3D lowers mean surface Chamfer from 0.229 to 0.223 m with all Actors and hazards retained (Table S14). Physical branches have identical action outcomes by construction: this validates

interface composition, not closed-loop benefit from repair.

H.3. V7.1 State Composition and Evidential Learning

Rigid dynamic–static composition audit. We identity-match 12 scene-0230 Actors (7 hazardous) between the frozen physical model and a read-only StreetGS checkpoint. Eleven are moving, and the retained trajectories contain 1,677 valid frames. Across fixed first/middle/last samples (36 Actor–frames), 5,791 physical Gaussians are transported by the checkpoint SE(3) poses while 106,807 appearance-owned and 1,140,862 Background Gaussians remain outside the physical query. The maximum inverse-query energy residual is 3.40×10^{-14} and the maximum pairwise-distance residual is 1.24×10^{-14} m, despite 126.28 m maximum translation. A first implementation that formed float32 pairwise distances directly in world coordinates produced a spurious 3.125 cm residual; we retain it as a numerical failure and use the mathematically equivalent inverse-query form, without changing geometry, cohort, or tolerance. This audit verifies deployment composition, not a new surface-accuracy or safety gain.

The geometry-locked appearance bridge assigns same-Actor StreetGS SH/opacity to all 5,791 carriers (mean Actor-level association distance 0.127 m) without copying visual geometry. Its physical surface improves Chamfer by 7.16 mm and hit recall by 3.19 points in this scene, but worsens all/hazard early returns by 1.60/0.60%. We therefore treat it strictly as evidence that image-trained attributes can be attached without changing geometry, not as a favorable appearance or physical example.

We then execute the bridge with the native StreetGS rasterizer on a view frozen solely by 3D frustum visibility (Fig. S5). Replacing the selected Actor’s 32,522 visual primitives with 309 physical carriers preserves the renderer interface but reduces footprint PSNR from 28.59 to 17.22 dB (-11.37 dB) and full-image PSNR by 3.99 dB. The blur and ghosting reject naive SH/opacity transfer as an appearance solution. We do not tune scale, opacity, primitive count, or view after observing this result, and the visual failure does not alter the GT-supervised physical carrier.

To separate a missing learning signal from insufficient carrier capacity, we next freeze every center, scale, rotation, trajectory, Background parameter, and non-target Actor attribute, and optimize only the selected Actor’s SH coefficients and opacity for 320 steps on eight geometrically preselected views. On six disjoint held-out views, footprint PSNR rises from 15.86 to 16.94 dB; all six views improve. The corresponding original StreetGS Actor reaches 25.35 dB, leaving an 8.41 dB gap. Thus image supervision can train attributes without changing physical geometry, but appearance-only optimization does not rescue the 309-primitive one-carrier representation. We do not convert

this single-scene capacity probe into an appearance or generalization claim.

We finally derive nine fixed tangent surfels per physical carrier using only an 8-neighbor PCA frame, the supervised carrier radius, and a 2 cm normal thickness. These 2,781 visual-only primitives never enter the physical query and their geometry receives no image gradient. Held-out PSNR reaches 17.57 dB, 0.63 dB above the one-carrier result, but the gap to the original remains 7.78 dB. Moreover, only the largest-footprint held-out view improves over the one-carrier model; the other five regress by 0.10–0.31 dB. A fixed coarse–fine hierarchy then combines the 309 parents and 2,781 surfels, assigning half the optical depth to each level before attribute-only refinement. It reaches 17.73 dB and improves all six views over the one-carrier result (median +0.47 dB, range +0.24 to +1.07 dB). These views were exposed before designing the hierarchy, so this is mechanism evidence rather than a fresh appearance benchmark; the 7.62 dB residual gap remains substantial.

We also test whether average geometry losses hide safety-relevant tails. Starting from M8, M29 adds a GT-only bidirectional loss equal to the mean of the worst 10% nearest-point distances, normalized per Actor, while retaining all set, plane, scale, frame, and smooth expected-depth losses. On the 66 exposed development Actors, target-to-surface tail distance improves by 0.924 mm, frame-mean distance by 0.843 mm, and Chamfer by 0.418 mm relative to M8. However, surface-to-target tail distance worsens by 0.362 mm and hazardous early-return rate rises from 26.37% to 26.69%. The geometry and ray gradients conflict in 86.6–90.6% of batches. This paired asymmetry rejects a single symmetric tail objective as a joint solution: increasing GT coverage can still move a small set of generated surfaces into unsupported free space. We stop without sweeping the tail fraction, weight, or seed.

M30 then preserves this conflict as the return interval in Eq. S17: immutable build anchors define the known-supported endpoint and anchors plus all M8 children define the possible endpoint. Set ordering has zero violations over 99,208 rays. Yet only 56.69% of GT returns are empirically bracketed, 47.03% of upper endpoints are unbounded, and the Actor-mean finite-width q90 is 0.326 m (0.343 m for hazardous Actors). More fundamentally, anchor-only queries already produce 19.58% early returns. The interval is therefore an honest epistemic interface, but not yet a tight collision certificate; the residual failure precedes completion and points to cross-frame canonical supervision or pose/visibility inconsistency.

M31 replays the exact 66 Actors through their original raw or processed producer and labels every anchor first return by construction provenance. It exactly recovers the 19.58% anchor early-return rate. Of 19,424 early rays, 66.70% terminate first on a kept query endpoint and 33.30%

on a projected endpoint. Projected points form only 13.36% of anchors, so they are overrepresented among contradictions by a factor of 2.49, but removing them cannot address the kept majority. The existing evidence rule changes raw-query early returns from 18.25% to 17.62%; rejected query points alone account for only 1.29% of rays. Hazardous versus clear rates are 20.02% versus 17.43%, while moving versus quasi-static rates are 19.63% versus 19.26%, and the target-frame trend is non-monotonic. Thus neither completion, projection, UNKNOWN filtering, nor motion alone explains the hard-anchor contradiction. We close post-hoc anchor filtering and instead treat free/occupied/unknown ray evidence as supervision of physical authority.

We next freeze every M8 center and scale and train only a continuous FREE/OCCUPIED/UNKNOWN mass per Gaussian from primitive ray votes and the same whole-ray categorical target. This removes both M19’s decoder compensation and M20’s scale shortcut, and occupied mass weights Gaussian energy continuously without thresholding or deletion. The model reduces evidential cross-entropy from 0.963 to 0.913, but changes categorical NLL by only 0.003. Relative to unit authority, all/hazard/clear early returns worsen by 0.336/0.338/0.322 points; hit recall improves by 0.487 points. Predicted mean anchor mass is $(m_F, m_O, m_U) = (0.127, 0.504, 0.369)$, indicating a learned global prior rather than primitive-specific contradiction resolution. We reject this geometry-only authority head without tuning. M31’s missing source-frame and multiview evidence must be retained by the corpus producer before authority can be identifiable from build-time inputs.

M33 performs this producer-side repair without changing the existing corpus. Exact replay aligns all 1,004 Actors and 297,535 anchors; 13.75% are projected. Build and held-out occupied masses have correlation 0.514, while direct three-state argmax agreement is only 43.41%, so build evidence is informative but is not itself an authority label. Moreover, 58.72% of anchors receive both free and occupied held-out votes, and projected anchors have higher mean held-out occupied mass than kept anchors (0.466 versus 0.425). These data reject provenance deletion and motivate a supervised continuous mapping from producer evidence to frozen Gaussian authority.

M34 learns that mapping while freezing every center, scale, trajectory, and completion-child authority. The predicted anchor occupied mass correlates 0.462 with held-out mass, establishing that the producer evidence repairs M32’s identifiability failure. Nevertheless, weighting the spatial Gaussian energy and normalizing it across ray samples increases all/hazard/clear early returns by 0.223/0.177/0.447 points and reduces hit rate by 0.753 points. This separates two failure levels: the model can predict which anchors receive occupied evidence, but the normalized energy CDF is not an ordered first-hit process. We reject M34 without tun-

ing and replace this composition with prefix transmittance rather than changing the supervision or deleting anchors.

M35 trains the same evidence head through analytic segment integration and prefix transmittance. Learned authority moves the ordered operator in the right direction: relative to unit transmittance, early returns fall from 42.29% to 35.27% and hits rise from 50.94% to 56.71%. However, both remain far behind the original unit-energy baseline of 18.19%/62.28%; hazardous early returns are 18.77 points worse. Unit and learned residual no-return masses are 0.320 and 0.399 despite conditioning evaluation on an Actor-box return. The failure is therefore optical parameterization, not ordering: summing unit line integrals over a dense, overlapping primitive set makes opacity depend on sampling density. We reject the current transmittance field and next separate anchor and completion-child optical contributions before defining a GT-calibrated surface measure.

The frozen M36 decomposition localizes that measure error. Learned anchors alone yield 18.39% early returns, only 0.20 points above the original baseline, whereas unit-opacity completion children alone yield 37.46%. Combining those children with learned anchors raises early returns to 35.27%. Thus a global optical scale or further anchor tuning would target the wrong component: the immediate missing supervision is completion-child FREE/OCCUPIED/UNKNOWN authority from its parent build evidence, kept separate from the observed-anchor head.

M37 supplies exactly that separate supervision while freezing every center, scale, trajectory, and the M35 anchor head. The child occupied mass correlates 0.431 with held-out ray evidence. Relative to unit-opacity children, learned child authority reduces early returns from 35.27% to 23.92% and increases hit recall from 56.71% to 65.59%. This is strong evidence that completion opacity is learnable from producer-side context rather than a post-hoc rule. It is not yet a physical pass: relative to the original unit-energy reference, all/hazard/clear early returns remain higher by 5.73/6.35/2.69 points, even as hit recall improves by 3.30 points. The ordered categorical NLL is almost flat. We therefore reject M37 and identify a loss-level omission: the observed interval before a LiDAR first return must itself supervise survival, rather than being represented only indirectly by competition with the return bin.

M38 adds that native supervision as the negative log survival contributed by completion children before the observed return, with all geometry and the anchor head frozen. It lowers held-out pre-hit child optical thickness by 32.0%, reduces early returns by 2.41 points relative to M37, and simultaneously raises hit recall by 1.33 points. Nevertheless, early returns remain 3.32 points above the original energy reference. The child evidential cross-entropy worsens from 0.934 to 1.058 as mean occupied mass falls from

0.481 to 0.383. Hence the supervision is useful, but the parameterization couples two incompatible roles: one scalar must preserve endpoint termination while also suppressing the isotropic Gaussian’s front tail. We reject M38 without changing its margin or weight and next isolate composition from authority.

That frozen M39 isolation resolves the apparent contradiction. We keep the M38 masses and all Gaussian geometry fixed, but interpret their weighted energy as a directly normalized categorical surface-return distribution instead of additive optical thickness. Relative to the original unit-energy reference, all/hazard/clear early returns decrease by 0.52/0.62/0.03 points, while hit recall increases by 2.17/2.32/1.46 points. All three preregistered criteria pass. An M37 descriptive arm has 0.05 points fewer all-stratum early returns but 1.57 points lower hit recall; it was not eligible for post-result selection. This experiment supports a precise representational conclusion: learned FREE/OCCUPIED/UNKNOWN mass is surface-return evidence under a categorical ray measure, not a volumetric opacity that can be summed across an arbitrarily dense primitive discretization. The result remains development-exposed because the frozen cross-domain test rejects transfer.

Directly fine-tuning both authority heads through that deployed distribution does not improve the candidate. M40 raises anchor/child occupied-mass correlation to 0.494/0.616 and adds 0.69 points of hit recall over M39, but early returns rise by 1.13 points. The completion-child mean occupied mass increases from 0.383 to 0.663, while each parent has four children. Thus the ray objective exploits a family-amplitude degree of freedom: it transfers total surface measure to the denser completion family. We reject the checkpoint despite its better evidential ranking and next conserve the separately supervised anchor and child measure before allowing within-family redistribution.

M41 enforces that conservation to within 1.53×10^{-5} per Actor, yet fine-tuning still raises early returns by 1.28 points over M39 while adding 1.07 points of hit recall. Anchor/child evidence correlations improve to 0.524/0.613. The remaining mismatch is therefore not total measure or identifiability: maximizing a nominal target-bin likelihood does not directly control whether the deployed CDF median crosses the early-return boundary. This motivates supervision on the ordered probability events used by the physical evaluation itself.

M42 consequently optimizes two GT-defined interval events under the same conserved representation: survival before $d_{\text{gt}} - 0.20$ m and return within ± 0.20 m of the observed endpoint. Aggregate early returns are 0.05 points below the unit-energy reference and hit recall is 2.54 points higher; hazardous early returns also improve by 0.11 points. However, clear-stratum early returns increase by

0.22 points, and the predicted early-event probability itself rises slightly during training while only the hit-band likelihood improves. We reject M42 under the frozen worst-stratum rule. This negative result is informative: defining physical consistency from GT events is necessary, but a global interval objective does not by itself remove stratum-dependent ambiguity in a fixed geometry and surface measure. We therefore stop development-set loss tuning and retain the simpler frozen M39 representation for one cross-domain evaluation, which later rejects transfer: all/hazard/clear early returns change by +0.23/ +0.54/ -0.04 points while hit recall changes by +5.89/ +6.30/ +5.54 points over 352 Actors and 1,016,652 rays. Only the hit-retention gate passes; no AV2 adaptation follows.

H.4. Categorical Responsibility, Visibility, and Temporal Diagnostics

M44 then applies the exact accounting in Eq. S26. Across all 99,208 rays, the CDF condition and rendered early decision agree exactly, and anchor-child decomposition has maximum residual 2.98×10^{-7} . M39 lowers mean pre-boundary mass by 0.331 points relative to unit energy: observed anchors contribute -0.346 points, whereas completion children contribute +0.016 points. For clear Actors, child mass instead rises by 0.163 points and cancels much of the 0.265-point anchor reduction. Thus the supported aggregate change is not a completion filter: it is driven by supervised anchor authority, while the remaining clear-stratum sensitivity is localized to child surface measure. Early and non-early rays have mean boundary CDF 0.827 and 0.114, respectively, furnishing a direct explanation of safety margin without creating a new deployment gate.

Finally, M45 replaces only the completion-child isotropic kernel by the frozen M11 GT-supervised oblate kernel. Relative to M39, all and clear early returns fall by 0.02 and 0.28 points, and hit recall rises by 3.87 points; the mean anisotropy is 8.10 with 0.020 m normal thickness. Hazardous early returns nevertheless rise by 0.034 points, so the frozen worst-stratum criterion rejects M45. This isolates a train-renderer mismatch rather than refuting planar support: M11 normals were learned for earliest hard intersection, while M45 uses them inside a global categorical CDF. We do not tune thickness; any successor must supervise orientation through that deployed CDF while freezing geometry and authority.

M46 does so, updating only the normal/thickness output rows with GT not-early and hit-band events. Training safe and hit NLL both decrease, and relative to M39 the final model lowers all/clear early returns by 0.05/0.59 points while adding 4.32 points of hit recall. Hazardous early returns instead increase by 0.062 points—more than the frozen M45 increase. We therefore reject and close oriented-support training. Renderer alignment is not suf-

ficient when a local support degree of freedom must also absorb motion-, pose-, or incidence-dependent global ordering; the next analysis keeps these factors separate rather than feeding the hazard label to geometry.

M47 separates those factors without further training. The small aggregate early reduction hides 2.996 points of newly early rays and 3.044 points of repaired rays. Motion is not the explanation: hazardous moving Actors improve by 0.015 points, whereas hazardous quasi-static Actors regress by 0.809 points, largely from one trailer and one truck. Prior KEEP/PROJECT provenance does not split the failure. Grazing and oblique rays improve by 0.832/0.483 points, while near-normal rays regress by 0.887 points; nevertheless, actor-level correlation between incidence and the early delta is only 0.052, and all five tested physical associations have absolute correlation at most 0.100. We therefore reject motion masks, provenance filters, and incidence thresholds. M45–M47 instead expose heterogeneous ray-label transport: local support is being asked to represent canonical shape, visibility, and termination ordering simultaneously. The next representation makes visibility an explicitly supervised factor of the joint return measure; temporal deformation remains a separate trajectory-supervised problem.

M48 tests that separation once by freezing M11 geometry and learning only a bounded child-visibility factor inside the same GT-supervised return measure. It fails more strongly: relative to M39, all/hazard early returns rise by 0.347/0.483 points, while clear rays improve by 0.322 points; hit recall rises by only 1.91 points and remains 1.95 points below the unweighted oriented M45 model. Training safe NLL decreases, yet predicted early probability increases and mean held-out visibility falls to 0.598. Thus merely moving attenuation inside the supervised renderer does not make it safe. Because the categorical distribution is normalized jointly, suppressing a component can move the boundary CDF in either direction depending on that component’s depth profile. We close visibility learning and next state this sign boundary analytically rather than tuning another head.

M49 verifies Eq. S27 on all 99,208 development rays. Uniform child attenuation has the adverse sign on 58.85% of rays and the safe sign on only 30.96%; under M48’s componentwise attenuation, adverse pressure dominates 65.45% overall and 67.27% for hazardous Actors. Accordingly, the exact boundary CDF rises on 59.32% of rays and falls on 27.27%. The first-order derivative predicts the sign of the finite change on 95.73% of rays, with mean exact/linear changes of +0.00513/ +0.00295. The observed 0.366-point early regression relative to M45 is the difference between 1.258 points of newly early and 0.892 points of repaired rays. This converts a negative ablation into a reusable boundary: component attenuation cannot provide

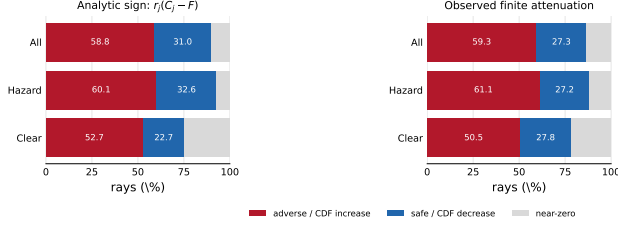


Figure S4. M49 attenuation sign boundary on 99,208 frozen rays. Red denotes the analytically adverse direction (left) or an observed increase in the pre-boundary CDF (right); blue denotes the corresponding safe/decreasing direction. The result is descriptive and is never converted into a selection rule.

a universal safety guarantee in a jointly normalized return measure.

M50 asks whether temporal physical consistency instead fails because pooled ray sampling underweights sparse target frames. Starting from frozen M8, it keeps the same canonical geometry and losses but averages smooth expected-depth and pre-hit hinge equally over target sensor origins. The four-epoch run improves Chamfer by 0.871 mm and changes all-ray hit recall by only +0.023 points, yet actor-mean worst-frame early rate rises by 1.135 points overall and 1.286 points for hazardous Actors. Aggregate all/hazard/clear early rates rise by 0.545/0.590/0.328 points. Geometry and physical gradients conflict in 83.2–86.6% of batches. We reject the single run without changing frame groups or weights: balancing GT frames changes the sampling measure but does not close the smooth-depth versus hard earliest-return surrogate gap, and it is not evidence for a motion or visibility head.

M51 pairs every M8/M50 ray to localize that gap. Hold-out smooth absolute depth error changes from 0.291706 to 0.292153 m, but mean loss alone is not the key result. On exactly the limited-anchor, non-voxelized training support, hard early rate rises by 0.490 points; 38.03% of newly early rays nevertheless improve smooth absolute depth error. On the formal full-anchor, 0.06 m voxelized surface, hard early rises by 0.545 points and the corresponding contradiction is 36.54% (33.97% for hazardous and 56.76% for clear Actors). Smooth-error change and hard-depth change correlate only 0.038. Deployment support realization flips 1.280% of events, but the same-support result proves that discretization is not the root cause. Expected depth and minimum supported depth require different guarantees; this pairing is explanatory and never used to select rays.

Qualitative protocol. Main-paper and supplement scenes are frozen from metadata and registered failure strata before rendering method outputs. Panels show RGB, LiDAR, physical surface, ray evidence, compiler action, and the re-



Figure S5. Frozen M24 view, cropped identically. Left to right: target image, original 32,522-primitive StreetGS Actor, and the 309-primitive geometry-locked carrier. Attribute transfer is rasterizer-compatible but loses appearance capacity; geometry is not moved to repair the image.

paired world. We report all frozen cases, including failures, rather than selecting only visually favorable examples.

I. Restored V7 Certificate and Visual Archive

This section restores the previously uncompiled V7 supplementary record. Cohorts, operators, and results retain their historical V7 identity. All table and figure references are local to this archive.

I.1. SceneIR Schema and Coordinate Conventions

SceneIR records explicit $T_{\text{city} \leftarrow \text{ego}}$, $T_{\text{ego} \leftarrow \text{sensor}}$, and $T_{\text{city} \leftarrow \text{actor}}$ transforms. A return $x_{s,t}$ is fused in Actor coordinates as

$$x_i = T_{\text{city} \leftarrow \text{actor}, t}^{-1} T_{\text{city} \leftarrow \text{ego}, t} T_{\text{ego} \leftarrow \text{sensor}, t} x_{s,t}. \quad (\text{S35})$$

The physical layer stores Actor-local surfels, known/unknown masks, normals, temporal and view support, matched rays, and source timestamps. Appearance primitives are referenced only through attachment provenance; Gaussian opacity is never read as collision occupancy. Actor identity, category, extent, trajectory, lifecycle, and analytic hazard fields are immutable across compiler actions.

Four actions. KEEP preserves supported query evidence. PROJECT moves an artifact only to its matched observed termination. COMPLETE adds a canonical surfel only when temporal and view support satisfy the frozen contract. UNKNOWN is the fallback when evidence is insufficient; it never becomes free.

I.2. Matched-Ray Physical Certificate

Automotive LiDAR is 2.5D ray evidence, not merely an unordered 3D point set [4]. For origin o , direction v , and observed termination d , the measured free interval is

$$\mathcal{F}(o, v, d) = \{o + sv \mid 0 \leq s < d - \epsilon\}. \quad (\text{S36})$$

A provenance-bearing PROJECT candidate is placed at the same ray’s termination $o + dv$. It therefore cannot lie in $\mathcal{F}$.

A candidate without an exact ray is marked UNKNOWN; nearest-surface projection is not substituted. This is the source of the paired zero-free-space result. It is a deterministic contract for matched rays, not a completeness proof for the whole Actor surface or a probabilistic sensor-noise guarantee.

The held-out targets are later frames excluded from both canonical fusion and query compilation. Depth, ray-termination, Chamfer, precision, and recall are computed against these target-only returns. This separation explains why zero paired free-space violation can coexist with the observed 17.19% per-Actor Chamfer tail.

Bidirectional visibility partition. For each held-out target ray, the nearest output within the frozen 0.20 m beam tube is classified as early, hit, or late; otherwise it is unmatched. In the reverse direction, an output point is supported at the first return, contradicted if it lies in the observed free interval, occluded if it is behind the return, and unknown when off ray. Occluded and unknown points are not penalized as free-space violations, preventing unseen Actor backs from being conflated with measurable contradictions.

The pooled result does not imply Actor-wise dominance: 64.04% add no early/visible contradiction and only 1.58% have absolute zero visible contradictions. All-Actor output attribution assigns 95.12% of new early rays to COMPLETE, but each costs against 14.96 new matched hits; target early returns fall by 9,439 overall. Moreover, only 13.49% of contradicted output points are completion points, so source-ray carving cannot selectively remove the observed tail. We retain completion and require the independent repair-or-abstain head for Actor-level authority.

The exact-once fresh cohort independently preserves this boundary: only 62.72% add no visible violation, 2.29% have exact zero visible contradiction, and 19.50% worsen in Chamfer. Among the latter, just 16.67% retain a non-inferior visibility F-score. Fresh confirmation therefore strengthens pooled three-dimensional evidence while continuing to reject a universal Actor certificate.

Frozen authority–certificate alignment. An exact identity join to the frozen P4 decisions shows why abstention must name its target. At 77.25% coverage, selected Chamfer-worsening falls from 19.50% to 14.60%, yet selected visible-failure risk is 36.39% versus 37.28% for always-repair. Its one-sided 95% Wilson upper bound is 40.40%, which remains above the baseline point risk; safe-visible AUROC is 0.533, and abstention captures 24.62% of visible failures. P4 authorizes Chamfer-nonworsening repair but provides no finite-cohort confidence separation for bidirectional visibility.

Why a stricter provenance rule fails. If authority additionally requires zero COMPLETE outputs, every retained point has a KEEP or matched-hit PROJECT witness. Provenance-only coverage is 36.90%; after conjunction with P4 it is 23.71%. Dual visible failure falls to 20.16% (one-sided 95% upper 26.70%), but Chamfer worsening rises to 43.55% and hazardous-Actor coverage is only 3.52%. Completion-count unsafe-visible AUROC is 0.589. Thus observed provenance improves one physical axis while failing joint geometry and hazard authority; it cannot certify unseen views.

A targeted visibility head still fails joint authority. A frozen nuScenes-only head reaches safe-visible AUROC 0.753 on source test and 0.625 on AV2. Visibility-only AV2 selection has 11.63% risk at 8.22% coverage. Its P4 conjunction has 10.26% risk (95% upper 20.98%) but only 7.46% coverage, 35.90% Chamfer worsening, and 3.52% hazard coverage. The ranking transfers; the authority does not.

Defer-to-query changes the unit of risk. Every abstained Actor retains its query surface, so Actor and hazard semantics remain 100.0%. P4’s population introduced-visible failure/composite Chamfer gain is 28.11%/0.083 m; P6-C is 30.78%/0.089 m; and P4-and-visibility is 0.76%/0.0012 m. The provenance policy is 4.78%/-0.0009 m and is strictly dominated by query-only. Thus low conditional risk can coexist with a worse fallback world.

Hazard-stratified accounting. For each frozen stratum, composite gain equals coverage times selected gain, while introduced-visible mass equals coverage times selected risk (maximum numerical residual 5.55×10^{-17}). Hazardous Actors are 27.15% of the cohort but account for 42.86% of P4 introduced failures and 56.42% of its Chamfer gain. P4 hazard risk is 47.37% versus 46.48% for always-repair; 45 of its 48 removed failures are instead clear. P6-C repairs 99.30% of hazardous Actors at 46.81% risk. Conversely, P4-and-visibility’s zero observed hazardous failure covers only 3.52% of hazardous Actors. Thus hazard-state retention is not a hazard-stratum visibility certificate.

Fresh target-nearest action provenance separates two failure channels. Across 1,435,391 target rays, the compiled surface adds 1.348% proxy new-early returns and yields 14.51 proxy new hits per proxy new-early return. COMPLETE accounts for 94.70% of proxy new early returns and 99.94% of proxy new hits, whereas KEEP accounts for 95.68% of emitted-surface contradictions. Hazardous/clear proxy new-early rates are 1.912%/0.980%;

Table S15. Target-nearest proximity evidence on consumed and independently frozen fresh AV2 cohorts. Higher hit/precision/F-score is better; lower early termination is better. Each audit was frozen before its metrics were read.

Surface	Target-hit recall (%)	Early termination (%)	Visible precision (%)	Visibility F-score
Consumed clean query	49.69	3.395	99.61	0.663
Consumed corrupted baseline	55.61	3.589	98.38	0.711
Consumed HARP-3D	69.83	2.868	99.77	0.822
Fresh clean query	49.72	3.448	99.51	0.663
Fresh HARP-3D	68.99	2.931	99.68	0.815

their COMPLETE hit-to-early ratios are 13.67/17.62. P4 and P6-C hazardous rates remain $1.006 \times / 1.001 \times$ always-repair. Thus the selectors do not suppress the completion-driven hazardous-ray mechanism. PROJECT has zero emitted provenance because observed-hit projections collapse under KEEP-first voxel deduplication; this is not a causal zero-effect result.

I.3. Literal First-Return Correction and Monotone Boundary

The frozen AV2 attribution above selects the Euclidean-nearest output to each target point and then checks its projected depth. It is a target-aligned proximity diagnostic, not literal ray termination. Following raw-LiDAR occupancy evaluation [5], P20/P22/P23 instead take the minimum positive projected depth among every surface point inside the same frozen beam tube. Query and compiled surfaces use the identical operator.

$$d_S(r) = \min\{s > 0 : o + sv \in S_\epsilon\}, \quad \min \emptyset = \infty. \quad (\text{S37})$$

Proposition 1 (one-sided deletion certificate). Fix ray r , tube radius ϵ , target depth d^* , tolerance τ , and query surface Q . If $S' \subseteq S$, then $d_{S'}(r) \geq d_S(r)$. Consequently, both the early indicator and new-early relative to Q are non-increasing:

$$\begin{aligned} e_Q(S; r) &= \mathbf{1}[d_S(r) < d^* - \tau] \mathbf{1}[d_Q(r) \geq d^* - \tau], \\ e_Q(S'; r) &\leq e_Q(S; r). \end{aligned} \quad (\text{S38})$$

Proof. The feasible positive depths in S' are a subset of those in S ; the minimum therefore cannot decrease, including the empty-set case $\min \emptyset = \infty$. Thresholding by the fixed $d^* - \tau$ preserves the order, and the query factor is constant. $\square$

The premise covers deletion only. Projection, completion, point motion, ray/tolerance changes, and unobserved collision queries lie outside it. Even under deletion, the sole point supporting a target-depth hit may be removed, while the two directed terms of symmetric Chamfer need not share a direction. Table S16 shows this exact boundary on the

consumed nuScenes source-development cohort: all three frozen deletion policies reduce hazardous early returns, yet all pay surface utility. P19 is most efficient, removing 131 hazardous events for 71 hits and 0.0919 mm mean Chamfer (1.845 events/hit; 1426.1 events/mm). These ratios are descriptive, not collision or road-safety guarantees.

P22 applies the identical frozen operator to the already consumed AV2 cohort rather than relabeling the proxy. Across 523 Actors and 1,435,391 target rays, all/hazard/clear new-early rates change from 1.348/1.912/0.980% to 9.894/13.926/7.256%, or $7.34/7.29/7.41 \times$. Every one of the 142,022 literal new-early returns has COMPLETE provenance; completion also supplies 97.843% of literal new hits, yielding only 1.194 hits per early return. This corrects, rather than confirms, the larger target-nearest efficiency ratio reported above.

P23 prospectively freezes the same comparison on a disjoint 10-log AV2 cohort before any quality read. Across 233 Actors and 650,145 rays, all/hazard/clear rates change from 1.370/1.788/0.948% to 10.801/14.817/6.750%, or $7.89/8.29/7.12 \times$. All 70,220 literal new-early returns are COMPLETE, and the hit/early ratio remains only 1.237. This is fresh confirmation of metric undercounting and provenance, not of a model or policy.

I.4. Frozen Cohorts and Exact-Once Rules

After a quality read, scenes/logs cannot be replaced, failed cases remain in the denominator, and no threshold, radius, feature group, or checkpoint may change. Download/resource failures are reported separately from scientific failures. The fresh AV2 protocol is bound to the already frozen P6-C/P4 artifacts and is not changed by the rejected fresh nuScenes result.

The fresh AV2 read contains 523 Actors. P6-C/P4 repair AUROC is 67.62/65.48, coverage is 83.94/77.25%, and selective Chamfer is 0.178/0.184 m. This external reversal does not dominate P4: false repair is 12.43% versus 11.28%, and P6-C independently loses fresh-nuScenes AUROC and AURC. The complete evidence therefore retains P4 and rejects a domain-invariant sparsity-consistency claim.

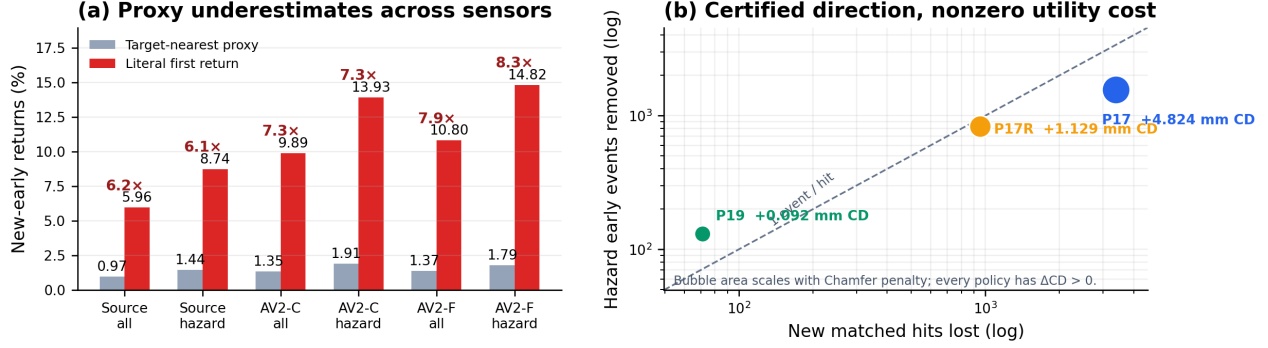


Figure S6. Frozen literal first-return boundary. (a) The target-nearest proxy understates total and hazardous-Actor new-early exposure on source, consumed AV2 (AV2-C), and exact-once fresh AV2 (AV2-F). (b) Deleting points is directionally safe for early returns, but every frozen source policy loses matched hits and worsens Chamfer. Bubble area scales with the Chamfer penalty. Ratios are consumed-source diagnostics; fresh AV2 transfers the metric failure only, not a model, policy, or road-safety bound.

Table S16. Literal first-return source audit. Hazard early is the fraction of held-out hazardous-Actor rays newly terminated before the target. Chamfer and new hits use the same frozen surfaces; no policy is refit.

Surface policy	Completion coverage (%)	Hazard new-early (%)	Mean Chamfer (m)	New hits
Always-complete	100.00	8.742	0.194587	32,575
P17 ray-only	71.43	7.777	0.199411	29,191
P17R ray+Chamfer	88.96	8.230	0.195716	31,620
P19 one-slot veto	98.95	8.662	0.194679	32,504

I.5. Actor-Level Decision Intervals

For the frozen ReLU validity MLP, an input box $[\ell, u]$ is propagated through each affine layer using

$$\ell' = W^+ \ell + W^- u + b, \quad u' = W^+ u + W^- \ell + b, \quad (\text{S39})$$

followed by endpoint-wise ReLU. Here $W^+ = \max(W, 0)$ and $W^- = \min(W, 0)$. The resulting logit interval is mapped through the sigmoid. An Actor is *robust-select* if its lower probability is above the frozen P4 threshold, *robust-abstain* if its upper probability is below threshold, and *unresolved* otherwise. This is deterministic interval-bound propagation over a specified box, related to robust-ODD certification [1]; it does not assert that the box describes an unseen sensor.

We fix radii 0.05/0.10/0.20 in nuScenes-training standard deviations. The sensor-opportunity group contains point/surfel counts, temporal/view support, range, and frame count; the physical-surface group contains surface residuals, keep/unknown fractions, completion fraction, and compiled/query ratio. Their union is exactly the 13D validity input.

At radius 0.10, 33.33% of nominal nuScenes-test selections and 91.23% of consumed-AV2 selections are robust-select. Nevertheless, 10.76% of the latter are retained target false repairs. Query-point count is the largest one-feature

interval source for 83.33%/95.11% of nuScenes/AV2 Actors. Thus a shifted model can be confidently and locally stable while wrong.

I.6. Intended Use and Safety Boundary

HARP-3D is intended for offline reconstruction audits: expose ray contradictions, compare Actor-preserving repairs, and report separate empirical task reliability. Consumers must retain unknown/provenance state and defer; repair never authorizes control. It is not a production collision map, closed-loop certificate or controller, or safety-case evidence. False repair can change collision support, sensor shift can cause confident errors, and selection can hide hazardous-Actor coverage. Deployment requires measured sensor errors, independent map/dynamics checks, closed-loop evaluation, and a domain-specific safety process. The interfaces expose, but do not remove, residual risk.

Table S17. Cohort construction is metadata-only. No quality, model score, RGB appearance, or target metric is read before freezing.

Cohort	Size	Frozen rule
AV2 consumed external	30 logs	Sort all 150 Sensor-val UUIDs; take indices 0, 5, . . . , 145. First 20 are quantitative and final 10 qualitative.
AV2 fresh exact-once	20 logs	Remove the consumed indices; from the 120-log complement take positions 0, 6, . . . , 114.
AV2 fresh metric exact-once	10 logs	Remove both prior AV2 cohorts; from the 100-log complement take positions 0, 10, . . . , 90, frozen before P22.
nuScenes fresh exact-once	20 scenes	Exclude every P4 train/calibration/test scene; sort the remaining 106 available scenes by official index and take rounded equally spaced positions.

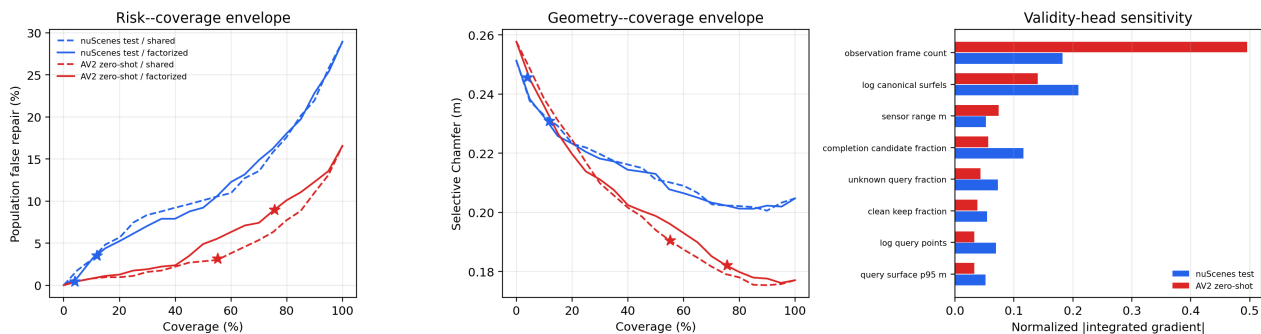


Figure S7. Complete frozen P7 explanation. Left and center: risk-coverage and geometry-coverage curves; stars mark the unchanged nuScenes-calibrated thresholds and never select an AV2 operating point. Right: Integrated Gradients relative to the nuScenes-training mean. The AV2 shift toward observation-frame count exposes sensor-opportunity dependence. The curves and attributions are descriptive, not a cross-domain certificate.

I.7. Failure Ledger and Claim Corrections

Table S18. Paper-facing failures are retained as boundaries rather than hidden by replacement experiments.

ID	Observation	Resulting boundary or correction
V7-F09	Aggregate geometry hides 17.19% Actor-level Chamfer worsening.	P3 is aggregate hard evidence; P4 adds repair-or-abstain and preserves all Actors.
V7-F11	AV2 validity attribution is dominated by sensor opportunity.	Zero hazard leakage does not imply domain-invariant validity.
V7-F13	Exact P4-train overlap is only two scenes/five Actors.	Reject joint fitting; audit the frozen physical/reliability interface only.
V7-F14	FP32 exceeds the analytic tolerance by 3.8×10^{-6} .	Recompute the same bound in FP64; do not loosen the tolerance or alter rows.
V7-F15	Fresh P6-C repair AUROC loses 3.50 points to P4.	Reject recovery despite higher operating-point coverage; P4 remains primary.
V7-F16	Freeze prose mislabeled the P346 held-out horizon.	Executable artifact is 2.5 s; correct documentation without rerunning.
V7-F17	47 AV2 false repairs remain robust-select at radius 0.10.	Network stability is not correctness, calibration, or a safety certificate.
V7-F18	Pooled ray physics improves, but consumed/fresh nonnew-visible rates are only 64.04/62.72%.	Retain aggregate C1 evidence and the completion gain–tail trade-off; reject a universal Actor certificate and do not tune the ray tolerance.
V7-F19	P6-C loses fresh-nuScenes ranking but improves fresh-AV2 AUROC while raising false repair.	Report the domain reversal; an external gate pass neither overrides the source rejection nor promotes P6-C over P4.
V7-F20	P4 visible risk is 36.39%, but its 40.40% upper bound exceeds the 37.28% baseline.	Repairability does not confidence-separate visibility risk.
V7-F21	Provenance lowers visible risk but raises Chamfer worsening to 43.55% and hazard-repair coverage falls to 3.52%.	Observed provenance is one-axis evidence, not joint authority.
V7-F22	The source visibility head transfers ranking, but dual coverage/gain are negligible and hazard-repair coverage is 3.52%.	Reject the head family rather than sweep target thresholds.
V7-F23	Provenance defer has -0.0009 m composite gain with nonzero introduced failure.	Query-only strictly dominates it; conditional risk is not fallback-system utility.
V7-F29	Literal first-return raises baseline hazard exposure to 8.742%; P17/P17R/P19 all lower it but worsen Chamfer.	Point deletion has a certified early-return direction, not hit, Chamfer, collision, or policy safety.

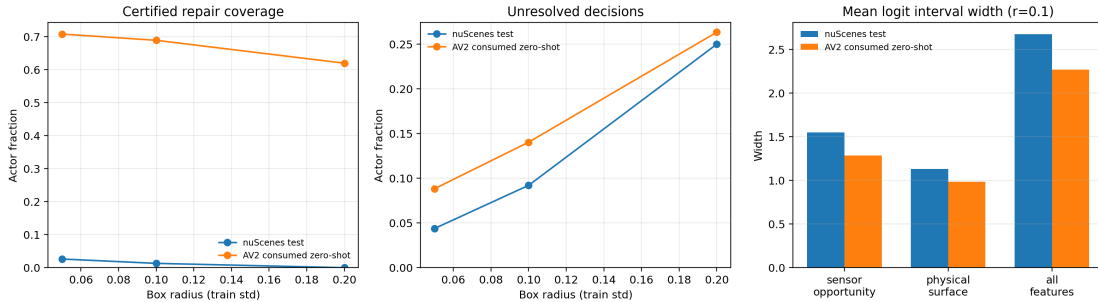


Figure S8. Frozen P7-C interval audit. Left/center: robust-select coverage and unresolved fraction across pre-registered radii. Right: sensor-opportunity boxes induce wider logit intervals than physical-surface boxes. The high AV2 certified coverage does not erase the 47 stable false repairs.

I.8. Frozen AV2 Camera-Evidence Gallery

The following ten cases are the actor-rank-0 case from each qualitative log in the pre-registered order, not a metric or appearance ranking. Every panel shows synchronized RGB, the paired synthetic artifact probe, the four-action physical output, and sparse camera depth. The full package contains 30 panels and 30 before/after videos (three lexically first eligible Actors per log). Videos diagnose temporal behavior but are not photorealistic renders.



Figure S9. Frozen case q00, hazardous Actor 1f197615, ring-front-center; 251 visible query returns and 2,287 sparse-depth points in crop.



Figure S10. Frozen case q01, non-hazardous Actor 0e0562b6, ring-rear-left; 137 visible query returns and 1,770 sparse-depth points.



Figure S11. Frozen case q02, non-hazardous Actor 08fc423e, ring-side-right; 85 visible query returns and 2,476 sparse-depth points.



Figure S12. Frozen case q03, hazardous Actor 05cccb41, ring-side-left; 2,456 visible query returns and 17,286 sparse-depth points.

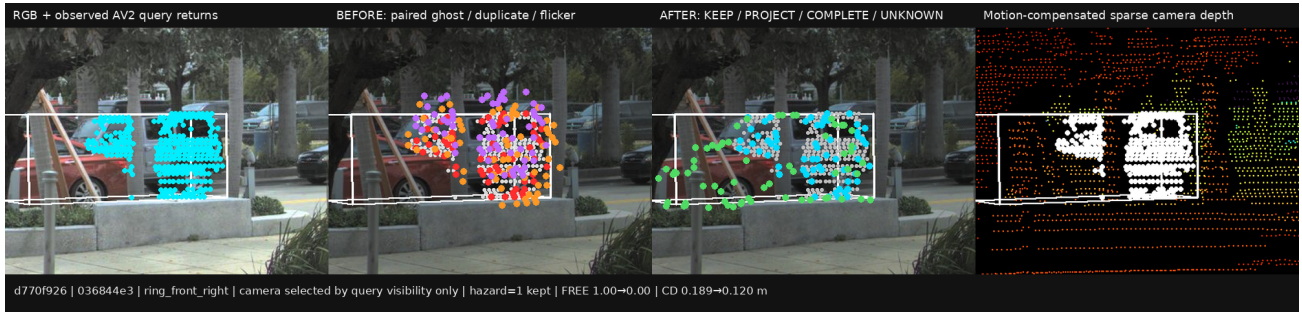


Figure S13. Frozen case q04, hazardous Actor 036844e3, ring-front-right; 292 visible query returns and 2,358 sparse-depth points.

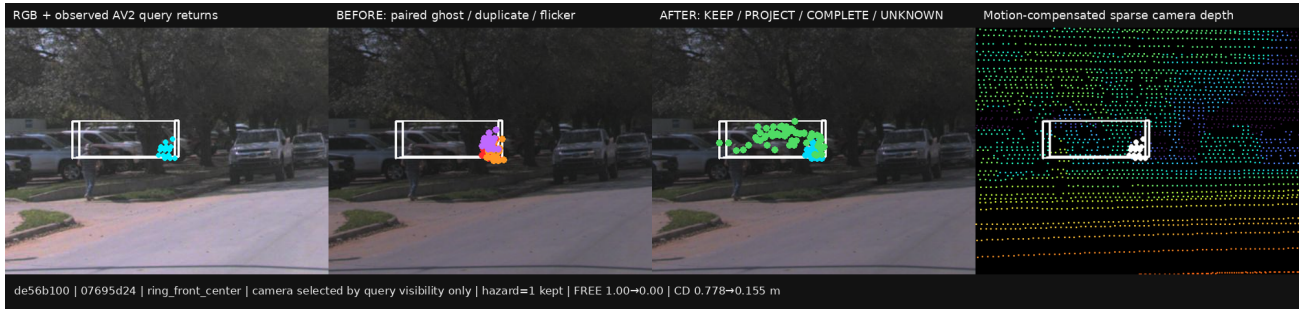


Figure S14. Frozen case q05, hazardous far/sparse Actor 07695d24, ring-front-center; all 14 query returns remain visible.



Figure S15. Frozen case q06, non-hazardous Actor 08515e0d, ring-front-right; 79 visible query returns and 2,757 sparse-depth points.



Figure S16. Frozen case q07, non-hazardous Actor 076c9103, ring-front-center. This unfiltered case exposes Chamfer worsening and is retained.

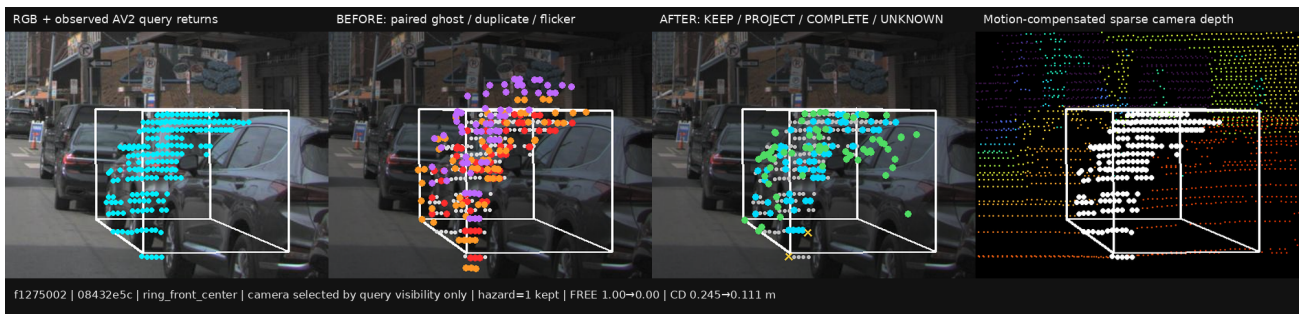


Figure S17. Frozen case q08, hazardous Actor 08432e5c, ring-front-center; 231 visible query returns and 2,101 sparse-depth points.



Figure S18. Frozen case q09, non-hazardous Actor 6d55ff95, ring-front-center; 38 visible query returns and 1,630 sparse-depth points.

I.9. Resource and Reproducibility Boundary

All V7 scientific runs use one RTX 3090. The largest paper-facing GPU allocation is below 1.5 GiB, while P7-C uses 9.2 MiB and 1.37 s. Download and preprocessing are sequential on the low-memory host. Artifacts are identified by task/run paths and immutable cohort/config files; the protocol intentionally does not add hashes, checksums, fingerprints, repeated smoke matrices, or post-hoc quality gates.

References

- [1] Julian Bitterwolf, Alexander Meinke, and Matthias Hein. Certifiably adversarially robust detection of out-of-distribution data. In *Advances in Neural Information Processing Systems*, 2020.
- [2] Loïc Chambon, Eloi Zablocki, Alexandre Boulch, Mickaël Chen, and Matthieu Cord. GaussRender: Learning 3D occupancy with gaussian rendering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.
- [3] Alec Farid, Sushant Veer, Boris Ivanovic, Karen Leung, and Marco Pavone. Task-relevant failure detection for trajectory predictors in autonomous vehicles. In *Proceedings of the 6th Conference on Robot Learning*, pages 1959–1969, 2023.
- [4] Peiyun Hu, Jason Ziglar, David Held, and Deva Ramanan. What you see is what you get: Exploiting visibility for 3D object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11001–11009, 2020.
- [5] Jonas Kälble, Sascha Wirges, Maxim Tatarchenko, and Eddy Ilg. Accurate training data for occupancy map prediction in automated driving using evidence theory. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5281–5290, 2024.
- [6] Hyeonseong Kim, Yoonsu Kang, Changgyoon Oh, and Kuk-Jin Yoon. Single domain generalization for LiDAR semantic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17587–17598, 2023.
- [7] Julian Ost, Fahim Mannan, Nils Thuerey, Julian Knodt, and Felix Heide. Neural scene graphs for dynamic scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021.
- [8] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 3319–3328, 2017.
- [9] Chinmay Talegaonkar, Yash Belhe, Ravi Ramamoorthi, and Nicholas Antipa. Volumetrically consistent 3D gaussian rasterization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10953–10963, 2025.
- [10] Adam Tonderski, Carl Lindström, Georg Hess, William Ljungbergh, Lennart Svensson, and Christoffer Petersson. NeuRAD: Neural rendering for autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [11] Ze Yang, Yun Chen, Jingkang Wang, Sivabalan Manivasagam, Wei-Chiu Ma, Anqi Joyce Yang, and Raquel Urtasun. UniSim: A neural closed-loop sensor simulator. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1389–1399, 2023.
- [12] Han Zhou, Jordy Van Landeghem, Teodora Popordanoska, and Matthew B. Blaschko. A novel characterization of the population area under the risk coverage curve (AURC) and rates of finite sample estimators. In *Proceedings of the 42nd International Conference on Machine Learning*, pages 79226–79253, 2025.